

**BURNOUT DETECTION FROM STUDENT-
GENERATED TEXT USING TRANSFORMER-BASED
NLP WITH PHRASE-LEVEL EXPLAINABILITY**

INDIVIDUAL THESIS

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Individual component: Emotional Burnout Detection from Student Text
with Phrase-Level Explainability

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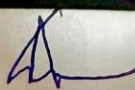
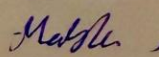
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Students often show emotional strain in what they write long before it appears in their grades. This thesis presents the emotional burnout detection component of R26-IT-059, an explainable multi-modal system for early academic burnout and dropout risk detection. The component reads a piece of student text and places it in one of seven mental-health categories (Normal, Stress, Anxiety, Depression, Bipolar, Personality Disorder and Suicidal). It then turns the result into a risk level and a distress score, and shows the words that drove each prediction. Since labelled student writing is not publicly available, a public corpus of 51,068 labelled mental-health texts from social media was used in its place. Ten models in four groups were compared on the same test set of 7,659 texts: untrained baselines, classical machine learning, a BiLSTM and fine-tuned transformers. A soft-voting ensemble of fine-tuned BERT and RoBERTa gave the most balanced result, with 82.32% accuracy, a weighted F1 of 0.823 and a macro F1 of 0.789. Bootstrap confidence intervals and McNemar's test showed that it was significantly more accurate than BERT alone but level with RoBERTa alone, with its main gain in macro F1. Explanations come from BERT's self-attention and are shown as a word heatmap, and LIME and SHAP picked out the same distress words on example texts. A simulated weekly test showed that distress language can be separated from ordinary language with about 90% accuracy at any point in a semester. The component was deployed as a Flask REST API with an advisor dashboard, and its emotional score was passed to the team's fusion layer. The main contribution is a seven-class classifier in which every alert comes with evidence that a human advisor can read and check. Validation on consented student writing is the main next step.

Keywords: student burnout, mental health text classification, explainable AI, transformer ensemble, early warning

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LIST OF ABBREVIATIONS

Abbreviation	Meaning	Abbreviation	Meaning
AI	Artificial intelligence	LIME	Local Interpretable Model-agnostic Explanations
ANN	Artificial neural network	LKR	Sri Lankan rupee
API	Application programming interface	LMS	Learning management system
AUC	Area under the ROC curve	LSTM	Long short-term memory
BERT	Bidirectional Encoder Representations from Transformers	ML	Machine learning
BiLSTM	Bidirectional long short-term memory	MNLI	Multi-Genre Natural Language Inference
BPO	Business process outsourcing	NLI	Natural language inference
CI	Confidence interval	NLP	Natural language processing
CLS	Classification token	OULAD	Open University Learning Analytics Dataset
CORS	Cross-origin resource sharing	REST	Representational state transfer
CPU	Central processing unit	RF	Random Forest
CRF	Conditional random field	RoBERTa	Robustly Optimized BERT Pretraining Approach
CSV	Comma-separated values	ROC	Receiver operating characteristic
EU	European Union	SDG	Sustainable Development Goal
FNN	Feedforward neural network	SHAP	SHapley Additive exPlanations
GPU	Graphics processing unit	SLIIT	Sri Lanka Institute of Information Technology
GRU	Gated recurrent unit	SVM	Support vector machine
HR	Human resources	SWOT	Strengths, weaknesses, opportunities and threats
HTTP	Hypertext Transfer Protocol	TF-IDF	Term frequency-inverse document frequency
ICAC	International Conference on Advancements in Computing	UAT	User acceptance testing
ICD-11	International Classification of Diseases, 11th revision	VADER	Valence Aware Dictionary and sEntiment Reasoner
JSON	JavaScript Object Notation	XAI	Explainable artificial intelligence
LDA	Latent Dirichlet allocation	XGBoost	Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall project context

Academic burnout is a state of exhaustion, cynicism and low sense of achievement caused by long study pressure [1], [2]. It is linked to lower achievement [3], and mental-health problems are common among university students [4]. Universities usually notice these problems late, after grades fall or a student stops attending.

The R26-IT-059 project, Explainable Multi-Modal AI System for Early Academic Burnout and Dropout Risk Detection, addresses this problem with four components (Figure 4.1): Academic Trend Prediction (IT22916426), which predicts dropout risk from learning management system (LMS) activity; Emotional Burnout Detection from Student Text (IT22196392, this thesis); Behavioural Anomaly Detection (IT22215710), which finds unusual LMS behaviour; and the Meta-Integration and Meta-XAI Decision Layer (IT22253194), which fuses the outputs into one risk level and advisor action.

1.2 Individual component context

This thesis covers the emotional component. It reads a short piece of student text, such as a reflection or a weekly check-in, and returns one of seven emotional states (Normal, Stress, Anxiety, Depression, Bipolar, Personality Disorder and Suicidal), a risk level, a distress score from 0 to 100 and the words that led to the result. It runs as a REST API behind the Emotional Analysis tab of the team dashboard and passes an emotional stress score to the team's fusion layer.

1.3 Problem statement

Students often show strain in their words before it shows in their marks [5], but advisors cannot read every message. Current text models help only partly. Many give a yes or no answer that does not show how serious a case is [6], most give a label with no reason [7], [8], and their labels are often weak proxies [9]. Advisors need a tool that shows the emotional state, how serious it is and why, so that a person can offer help early.

1.4 Research gap

- The best reported accuracies come from binary tasks, which do not show how serious a case is [6], [10].
- Explanations, when given, describe a whole dataset rather than one prediction [6], [11], and the most accurate seven-class study gives none [12].
- Ensembles are rarely paired with an explanation of each prediction.
- Emotional signals in text are rarely followed over time or combined with academic and behavioural signals in one early-warning system.

1.5 Aim and objectives

Aim: to develop and evaluate an explainable natural language processing (NLP) component that detects emotional distress in student text across seven categories, explains each prediction at word level, and supplies a risk signal to advisors and to the fusion layer.

- O1. Prepare a clean seven-class text corpus and a fixed, stratified split.
- O2. Compare ten models from four groups under one evaluation protocol.
- O3. Fine-tune BERT and RoBERTa and combine them in a soft-voting ensemble.
- O4. Handle class imbalance and test a class-weighted loss.
- O5. Explain each prediction with word-level attention and check it against LIME and SHAP.
- O6. Turn each prediction into a risk level, a distress score and an emotional stress score.
- O7. Test whether the differences between the strongest models are statistically significant.
- O8. Deploy the component as a REST API and dashboard and integrate it with the team system.

1.6 Research questions or hypotheses

- RQ1. How well do fine-tuned transformers classify text into seven mental-health categories, compared with untrained, classical and recurrent baselines?
- RQ2. Does a soft-voting ensemble of BERT and RoBERTa give a real gain over the single models?

- RQ3. Do attention explanations highlight meaningful words and agree with LIME and SHAP?
- RQ4. In a simulated weekly setting, can distress language be separated from ordinary language early in a semester?

1.7 Scope limitations and assumptions

- The component works on English text and uses a public social-media corpus, because labelled student writing is not publicly available.
- The component detects emotional distress states linked to burnout. It does not diagnose.
- The weekly results come from a simulation, because the corpus has no time stamps.
- The uses beyond universities in Chapter 8 are proposals for future work.

1.8 Individual contribution

- A cleaned seven-class corpus, and ten models in four groups compared on one test set with confidence intervals and McNemar's tests.
- A BERT and RoBERTa soft-voting ensemble with word-level attention explanations, checked against LIME and SHAP.
- Risk levels, a distress score and an emotional stress score for the fusion layer.
- A simulated lead-time test.
- The Flask REST API (port 5004), the Emotional Analysis dashboard tab and the score export for the Meta-Integration layer.

1.9 Thesis structure

Chapter 2 reviews related work and Chapter 3 describes the methodology. Chapter 4 presents the design and Chapter 5 its implementation, integration and testing. Chapter 6 reports the results and Chapter 7 discusses them. Chapter 8 covers commercialisation and sustainability, and Chapter 9 concludes.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review began with the 22 studies examined in the author’s research paper and was then extended for this thesis. Google Scholar was the main search tool. IEEE Xplore, the ACL Anthology, PubMed Central, publisher websites and the reference lists of key papers were used to find further work. Search terms included “student burnout prediction”, “mental health text classification”, “stress detection social media”, “depression detection BERT”, “RoBERTa mental health”, “explainable AI NLP”, “attention explanation”, “early warning system higher education”, “employee burnout text” and “healthcare worker burnout”.

Applied studies published between 2013 and 2025 were included, together with the original papers of the methods used in this thesis. A study was kept if it (a) detected stress, burnout, depression or another mental-health state from text or student data, (b) explained the output of a model, (c) predicted student risk early, or (d) studied burnout or distress at work through language. Each claim cited in this chapter was checked against the original publication, and sources that could not be found, or whose content did not support the claim, were removed.

The selected work is presented in two parts. Section 2.2 covers the theory and methods that the component builds on. Section 2.3 groups the related research by theme rather than paper by paper, so that the direction of the field is easy to follow. It uses the five themes of the research paper and adds a sixth theme on burnout at work. Early studies used classical machine learning on records and surveys. Later work moved to transformer models that read text directly, and the most recent work has started to deal with explanation and timing. The main studies are compared in Table 2.1.

2.2 Domain and theoretical foundations

2.2.1 Burnout and student well-being

Burnout was first described in work settings as a combination of emotional exhaustion, cynicism and reduced professional efficacy [1]. Schaufeli et al. showed

that university students experience the same three dimensions in relation to their studies [2]. Student burnout matters for outcomes: a meta-analysis of more than 100,000 students found that burnout is linked to lower academic achievement [3]. Burnout also overlaps with common mental-health problems, which are frequent among university students [4].

Because burnout builds up over weeks and months, a useful early-warning system has to notice the emotional signs early and repeatedly, not only at the end of a course. This thesis does not measure burnout directly. It detects emotional distress states that commonly go with burnout, such as stress, anxiety and depression, and treats them as a proxy signal for an advisor to review (Section 1.7).

2.2.2 Language as a signal of mental state

The words people choose reflect how they feel. De Choudhury et al. showed that features of a person's social-media posts in the year before a reported depression diagnosis could be used to predict that diagnosis [5]. Coppersmith et al. found measurable signals of several mental-health conditions in Twitter posts [13]. Reviews of this field agree that language carries useful signals of depression, anxiety, stress and suicidal thinking [14], [15], [8].

The same reviews also point out weaknesses. Labels are often based on proxies such as the forum a post came from, the data comes mostly from social media, and models are rarely tested outside the dataset they were trained on [7], [9]. These warnings shape how the results of this thesis are framed: the output is a signal for human review, not a diagnosis.

2.2.3 From lexicons to transformers

Early text tools used sentiment lexicons. VADER, for example, scores a text with a hand-built list of words and grammar rules, with no training on the target task [16]. Classical machine learning improved on this by learning from labelled examples, usually with term frequency-inverse document frequency (TF-IDF) features, which weight each word by how often it appears in a text and how rare it is in the whole corpus [17]. These models treat a text as a bag of words, so they lose word order and

context. Recurrent networks such as the long short-term memory (LSTM) network read the words in order and keep a memory of what came before [18].

The transformer replaced recurrence with self-attention, which lets every word look at every other word in the text [19]. BERT pretrains a bidirectional transformer on large amounts of general text, and it can then be fine-tuned for a classification task by adding one small output layer [20]. RoBERTa keeps the same architecture but trains for longer, on more data, with dynamic masking and a byte-level vocabulary, and it usually performs better on later tasks [21]. Distillation produces a smaller student model that keeps most of the accuracy of the original [22].

Two further ideas are used as baselines in this thesis. In zero-shot classification, a model trained for natural language inference judges whether a statement such as “This text expresses depression” follows from the input, so no task-specific training is needed [23], [24], [25]. Emotion models are transformers that have already been fine-tuned to recognise basic emotions such as joy, sadness and fear [26]. Finally, combining several models in an ensemble often gives more accurate and more stable predictions than a single model, because the models make partly different errors [27].

2.2.4 Explainable AI for text

Explainable AI aims to make a model’s decision understandable to the people who use it [28]. Explanations can be local, explaining one prediction, or global, describing how the model behaves overall. They can come from inside the model or from a separate method applied after training [29]. LIME explains one prediction by fitting a simple model to the behaviour of the original model on many small changes of the input [30]. SHAP gives each input feature a share of the prediction based on Shapley values from game theory [31].

For transformers, the attention weights can also be read directly, at no extra cost when the prediction is made. Whether attention is a faithful explanation is debated. Jain and Wallace showed that attention weights do not always agree with other measures of importance [32], while Wiegreffe and Pinter argued that attention can still be a useful explanation when it is checked carefully [33]. This debate is the

reason why the attention explanations in this thesis are compared with LIME and SHAP (Section 3.3).

2.2.5 Design science

The study follows design science research, in which knowledge is created by building an artefact and evaluating it carefully [34]. Here the artefact is the explainable classification component, and the evaluation is a controlled comparison of models on a shared test set (Chapter 3).

2.3 Related research

The related research falls into six themes. Each theme is summarised below with its main limitation, and the key studies are compared in Table 2.1.

2.3.1 Traditional machine learning on administrative and survey data

The earliest work on burnout prediction used structured data, such as university records and questionnaires, with classical machine learning. A survey by Grządzielewska describes the statistical and machine-learning approaches that have been used to predict burnout [35].

Yeskuatov et al. tested a strict version of this idea. They asked whether burnout among undergraduate computing students could be detected from university records alone, so that no psychological survey would be needed to make a prediction. A burnout survey was used only to create the labels. The best model reached an F1 of only 68.4% for the exhaustion dimension [36]. Records that are collected anyway therefore carry some signal, but not enough to capture emotional states.

Studies that use psychological questionnaires reach much higher scores. Pereira et al. surveyed 274 psychology students, built burnout risk profiles, and predicted them with Random Forest, XGBoost and Support Vector Machine (SVM) models, with a best accuracy of 97.53% [37]. Larraín et al. combined Random Forest and neural network classifiers on questionnaire answers from 93 participants in Valparaíso, Chile, and detected signs of burnout, anxiety and depression with high accuracy and few false negatives [38]. Together, these studies show that a classical model is only as good as its input: when the features hold psychological information the results are strong, and when they do not, the results are weak.

A second group kept classical methods but changed the input from records to text. Merhbene et al. collected Reddit posts describing burnout and trained ensembles of logistic regression models on balanced batches of data. The best ensembles reached a balanced accuracy of 0.93 but an F1 of only 0.43 on the very unbalanced test set [10]. This gap between two common measures shows how strongly class imbalance can change the picture that a study gives.

Limitation. Survey data has to be collected on purpose, depends on honest self-report and gives a snapshot rather than an ongoing signal. Most of these studies also give a single label or score, and none of them explains an individual prediction to the person who has to act on it. Text that students already write offers a passive alternative, which is the route taken in this thesis.

2.3.2 Transformer-based approaches for student mental health

Before transformers, deep learning models were already used on mental-health text. Kim et al., for example, trained a separate model for each of six conditions on Reddit posts [39]. Transformers then brought a clear step up, because self-attention lets the model read every word in the context of the whole text (Section 2.2.3).

Several studies applied transformers to the stress and depression of students. Beegam and Baalaji fine-tuned BERT with a deep neural network on a Kaggle depression and anxiety text dataset of 7,489 posts and reached 97% accuracy, but the task was binary: depressed or not [6]. Zhuang et al. combined BERT with topic modelling and a sequence labelling model to detect the psychological stress of postgraduate students from social-media posts. They reached 92.55% accuracy and an F1 of 92.18%, and noted that irony and sarcasm in short posts are hard to detect [40]. Wan et al. combined RoBERTa with recurrent GRU layers to separate depressive from non-depressive posts, using about 13,000 Reddit posts and 6,164 tweets, and reported 90.18% accuracy. They also noted that the posts were not confirmed to come from college students [41].

Other studies used transformers to detect stress more widely. Nijhawan et al. used BERT for sentiment classification, together with topic modelling by latent Dirichlet allocation (LDA), on large sets of tweets to detect stress [42]. Ilias et al. added extra-

linguistic features to BERT and MentalBERT and used label smoothing, so that the confidence of the models for stress and depression detection was better calibrated [43].

The study closest to this thesis is by Khan et al. They classified a combined Kaggle mental-health corpus with the same seven labels used here (Normal, Depression, Suicidal, Anxiety, Stress, Bipolar and Personality Disorder) and reported 97% accuracy and an F1 of 0.94 with RoBERTa-Large [12]. Their study shows what a large model can reach on this kind of data, and that a seven-class task is possible.

Limitation. The strongest results come either from binary tasks or from large models. A binary label tells an advisor little about how serious a case is, and none of these studies gives the reason behind an individual prediction.

2.3.3 Domain-adapted models and the black-box problem

A separate line of work adapts models to the language of mental health, and public datasets made this possible. Dreaddit provides Reddit posts labelled for stress [44]. The eRisk collection orders each user's writings in time, so that systems can be rewarded for detecting depression early [45]. SMHD links Reddit users to nine self-reported mental-health conditions [46], and Gkotsis et al. classified posts into mental-health themes based on the subreddit they came from [47]. Murarka et al. used RoBERTa to classify such informal Reddit posts into several mental-health conditions [48]. The dataset used in this thesis is a compiled Kaggle corpus that merges several of these sources under seven labels [49].

Domain adaptation goes one step further and pretrains the model on the target language. Ji et al. released MentalBERT and MentalRoBERTa, which were further pretrained on mental-health posts, and reported gains on several detection tasks [50]. A related idea is to start from a model that already recognises basic emotions such as joy, sadness and fear [26]. This thesis tests such an emotion-pretrained model, Emotion-DistilRoBERTa, as one of its comparison points (Chapter 6).

The black-box problem runs through this whole line of work. As models become larger and more specialised, they report a label and a probability, but not the reason behind them. Beegam and Baalaji used word clouds of the whole dataset to show

which words were common in depressed texts [6], which describes the data but not one decision. Wan et al. describe their model as interpretable, but they do not show which words led to a given prediction [41], and Khan et al. report accuracy without an explanation for each text [12].

Limitation. Domain data and domain pretraining raise accuracy, but the labels in these datasets come from the forum a post was taken from or from self-report, not from a clinician [7], [9]. Models trained on them detect language associated with a condition, not the condition itself, and they rarely tell the user why a text was flagged.

2.3.4 Explainability attempts and ensemble strategies

As accuracy rose, two questions moved to the front: how to explain a prediction, and how to combine models without losing that explanation.

Some studies tried to open the box. Mazurets et al. split a user's posts into topics and showed a word cloud and a fatigue indicator for each topic, with a binary BERT classifier reaching 83% accuracy [11]. In educational prediction more widely, Nagy and Molontay used explainable AI tools on dropout predictions at a Hungarian university, with the aim of supporting personalised intervention for each student [51]. Kalita et al. paired a Bi-LSTM with SHAP so that predictions of students' grade point averages could be explained [52]. For transformers, the attention weights can be read at no extra cost, but they are not always a faithful explanation [32], so they should be checked against other methods [33].

On the model side, ensembles are a common route to higher accuracy. Combining several models often gives more accurate and more stable predictions, because the models make partly different errors [27]. Larraín et al. combined Random Forest with a neural network [38], and Merhbene et al. used ensembles of logistic regression models [10]. The cost is transparency: every model added to an ensemble makes it harder to say why the final answer was given.

Limitation. In the reviewed text studies, a per-prediction explanation that a non-technical advisor can read, and that has been checked against an independent method, was rare. Ensembles raise accuracy, but they were seldom paired with such

an explanation. This thesis therefore combines a BERT and RoBERTa ensemble with a word-level attention explanation and checks that explanation against LIME and SHAP.

2.3.5 Temporal prediction horizons and early intervention

Student-success research asks how early a reliable warning can be raised while there is still time to help. Akçapınar et al. showed that prediction accuracy rises as more weeks of a course are observed [53], which means that an earlier warning is usually a less certain one. Al-azazi and Ghurab combined an artificial neural network with an LSTM to predict student performance early in online courses [54]. OU Analyse, used at the Open University, gives teachers weekly predictions of which students are at risk of failing [55]. In mental-health text research, the eRisk collection was designed to reward systems that detect risk from as few early posts as possible [45]. Wan et al. used recurrent layers to follow the order of a user’s posts, but they did not track the emotional state of individual students week by week [41].

Choosing the prediction horizon is therefore a trade-off between how early a warning is raised and how reliable it is. For emotional signals in student text, this trade-off has hardly been studied.

Limitation. These studies show that timing matters, but they mostly use academic or activity data. Weekly emotional signals from student text are rarely studied, mainly because time-stamped and labelled student writing is not publicly available. This thesis looks at the question through a simulated lead-time test (Section 3.3).

2.3.6 Burnout and distress at work

Burnout was first studied in workers, and it remains mainly a workplace problem. The World Health Organization’s International Classification of Diseases (ICD-11) describes burn-out as a syndrome that results from chronic workplace stress that has not been successfully managed, with three dimensions: exhaustion, mental distance from one’s job, and reduced professional efficacy [56]. The same signs appear in students’ relation to their studies (Section 2.2.1).

Language studies show that work stress leaves traces in what people write. Li et al. analysed about 1.5 million tweets from nurses and construction workers between

2018 and 2022. Nurses, especially younger nurses, used more burnout- and stress-related words, and negative sentiment dominated their tweets from early 2020 [57]. Malgaroli et al. applied topic modelling to the therapy messages of 820 healthcare workers and 820 matched people from other jobs. In the healthcare workers, talk about work on the hospital unit, mood and sleep was linked with moderate to severe anxiety or depression [58]. The studies by Merhbene et al. and Mazurets et al. (Sections 2.3.1 and 2.3.4) also dealt with burnout in professional settings. In industry, staff survey platforms group open comments into topics with a sentiment label [59], and work-pattern tools point out habits linked to burnout, such as regular after-hours work and meeting overload [60].

Limitation. Workplace studies mostly describe groups after the fact or summarise comments by topic. None of the reviewed studies gives a graded emotional state for each text together with word-level evidence, and the privacy questions of using such signals at work are seldom addressed. These points shape the workplace use of the engine proposed in Chapter 8.

Table 2.1: Comparison of related work

Study	Data	Task (classes)	Method	Explains each prediction	Graded risk output
Yeskuatov et al. [36]	Administrative records	Burnout dimensions	Supervised ML	No	No
Pereira et al. [37]	Survey of 274 students	Burnout risk profiles	RF, XGBoost, SVM	No	Profiles
Larraín et al. [38]	Questionnaire, 93 participants	Burnout, anxiety, depression (present or not)	RF and ANN ensemble	No	No
Merhbene et al. [10]	Reddit posts	Burnout or not (2)	Logistic regression ensembles	No	No
Beegam and Baalaji [6]	Kaggle text, 7,489 posts	Depressed or not (2)	BERT with a deep neural network	Dataset word clouds only	No
Zhuang et al. [40]	Postgraduate social-media posts	Stress profile	BERT, LDA and BiLSTM-CRF	No	No
Nijhawan et al. [42]	Large sets of tweets	Stress, through sentiment and emotion	BERT and other ML models, LDA topics	Topic level only	No
Wan et al. [41]	About 13,000 Reddit posts and 6,164 tweets	Depressive or not (2)	RoBERTa with GRU layers	No method shown	No
Khan et al. [12]	Kaggle mental-health corpus	Seven labels	RoBERTa-Large	No	No

Study	Data	Task (classes)	Method	Explains each prediction	Graded risk output
Mazurets et al. [11]	Tweets of healthcare workers	Digital fatigue or not (2)	BERT with LDA topics	Topic segments and word clouds	Fatigue indicator
Li et al. [57]	1.5 million tweets of nurses and construction workers	Burnout and stress language (descriptive)	Computational linguistic analysis	No	No
Malgaroli et al. [58]	Therapy messages of 820 healthcare workers and 820 controls	Topics linked with anxiety and depression	Structural topic models	Topic level only	No
This thesis	Kaggle mental-health corpus	Seven labels	BERT and RoBERTa soft-voting ensemble	Yes: word heatmap, checked with LIME and SHAP	Four risk levels and a 0 to 100 distress score

2.4 Existing systems products and techniques

Several tools already flag students at risk, but none of them covers the need addressed in this thesis.

- **LMS analytics.** Moodle’s built-in analytics include a model that flags students at risk of dropping out, based on how deeply and how widely they take part in course activities. The result is a yes-or-no flag drawn from activity data, with no view of how the student feels [61].
- **Institutional predictive analytics.** OU Analyse gives teachers weekly predictions of students at risk of failing, based on demographic and virtual learning environment data [55]. Commercial student-success platforms such as EAB Navigate and Civitas Learning provide risk scores from academic and enrolment data.
- **Student safety monitoring.** In schools in the United States, tools such as GoGuardian Beacon, Gaggle and Bark scan students’ online activity on school devices and alert staff to possible self-harm, suicide or violence [62], [63]. Beacon uses AI-based detection and leaves the final judgement to school staff [62]. These tools are close to this component in purpose, but they work by monitoring school-issued devices, and they have been criticised for false alarms, privacy risks and alerts that sometimes reach the police [63].

- **Workplace well-being tools.** Staff survey platforms such as Workday Peakon Employee Voice group open comments into topics with a sentiment label [59], and Microsoft Viva Insights points out work habits that may lead to burnout [60]. They work on survey topics or work patterns, not on a graded emotional state for each text.
- **Off-the-shelf techniques.** Lexicon tools such as VADER [16], zero-shot classifiers [23] and general emotion models [26] can be used without training, but Chapter 6 shows that, without fine-tuning on this task, they perform poorly.

None of these gives a university advisor a seven-class emotional state for a piece of text, together with the evidence behind it and a risk level, inside a wider early-warning system. A detailed comparison with these tools is given in Table 8.2 (Chapter 8).

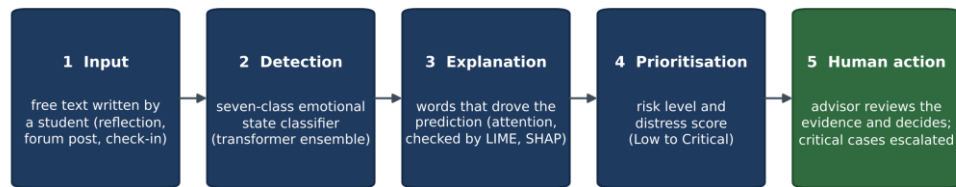
2.5 Research gap and conceptual framework

Taken together, the six themes show three clear trends. Transformer models read emotional signals in text better than classical methods, explanation is now seen as necessary rather than optional for any system used in education, and early, time-aware prediction is what makes timely help possible. Against these trends, four gaps remain:

- G1. The highest-scoring text models are mostly binary, so they do not show how serious a case is.
- G2. Explanations are missing or global, so an advisor cannot see why one particular text was flagged.
- G3. Ensembles raise accuracy but are rarely paired with a per-prediction explanation.
- G4. Emotional text is studied as single snapshots. How distress develops over a semester, and how early it can be noticed, is rarely studied, and the signal is rarely fed into a wider early-warning system.

None of the reviewed studies addresses all four gaps together. The conceptual framework of this thesis (Figure 2.1) links these gaps to the design. Student text goes into a seven-class detector (G1). Each prediction comes with word-level evidence

(G2). The detector is an ensemble with the explanation built in (G3). The result is turned into a risk level, shown across the weeks of a semester and passed to the team's fusion layer (G4). In every case a human advisor makes the final decision.



Gaps addressed (Section 2.5): G1 binary labels → seven classes · G2 opaque output → phrase-level evidence ·
G3 ensembles hide reasons → ensemble with built-in explanation · G4 isolated snapshots → weekly view and fusion input

Figure 2.1: Conceptual framework of the emotional component, from student text to human action

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows a design science approach [34]: an explainable text classifier was built and then evaluated with a quantitative experiment. Ten models, from untrained baselines to fine-tuned transformers, were compared on one held-out test set, and the differences between the strongest models were tested statistically. Figure 4.2 shows the full pipeline.

3.2 Data participants or study materials

No human participants were involved. The data is the public “Sentiment Analysis for Mental Health” corpus on Kaggle [49], which merges several public mental-health text datasets, mostly from Reddit and other social media, under seven labels (Table 3.1). Labelled student writing is not publicly available, so this corpus is used as a proxy.

The raw file has 53,045 rows. After removing 364 rows with missing values and 1,612 duplicate texts, 51,068 passages remained, and 51,055 were used after 13 became empty during cleaning. The classes are imbalanced: Normal has 16,039 passages and Personality Disorder only 895 (Figure 3.1).

Table 3.1: Definitions of the seven classes

Class	Meaning in this thesis
Normal	Ordinary, stable text with no sign of distress
Stress	Feeling under pressure, overwhelmed or unable to cope
Anxiety	Heavy worry, fear, nervousness or panic
Depression	Lasting sadness, hopelessness or loss of interest
Bipolar	Strong mood swings between high and low states
Personality Disorder	Lasting problems with identity, self-image or relationships
Suicidal	Thoughts of self-harm, a wish to die or intent to end one’s life

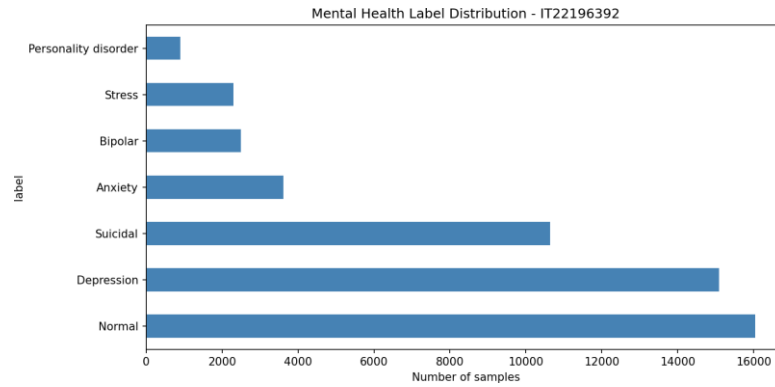


Figure 3.1: Number of passages in each of the seven classes after cleaning

3.3 Experimental or study procedure

Pre-processing and split. Each text was lower-cased, and links, user mentions, hashtags and non-letter characters were removed. The 51,055 passages were split 70/15/15 into training (35,738), validation (7,658) and test (7,659) sets with stratified sampling and a fixed seed (42). The corpus has no author identifiers, so the split was made at passage level.

Models. Ten models in four groups were built (Table 3.2). The classical models used TF-IDF features of single words and word pairs [17], the BiLSTM read word sequences of up to 150 tokens [18], and the transformers used their own tokenizers with up to 128 tokens [20], [21]. VADER [16] and a zero-shot BART-large-MNLI classifier [24], [23] were used without training; the zero-shot model was scored on 500 test passages because it is slow. The emotion-pretrained model is the public Emotion-DistilRoBERTa [26], which the project notebooks call “Mental-RoBERTa”.

Table 3.2: Overview of the ten models

#	Model	Group	Key settings	Parameters
1	VADER	Untrained (lexicon)	Compound score mapped to classes by thresholds	None trained
2	Zero-shot classifier	Untrained (pretrained NLI)	facebook/bart-large-mnli, one hypothesis per class	None trained
3	Logistic Regression	Classical ML	TF-IDF (10,000 features), balanced class weights	10,000 features
4	Random Forest	Classical ML	TF-IDF, 200 trees, balanced class weights	10,000 features
5	Support Vector Machine	Classical ML	TF-IDF, linear kernel, balanced class weights	10,000 features

#	Model	Group	Key settings	Parameters
6	BiLSTM	Deep learning	Two bidirectional layers, 128 units per direction	3,221,255
7	Emotion-DistilRoBERTa	Transformer	Emotion-pretrained DistilRoBERTa, fine-tuned	about 82 million
8	BERT	Transformer	bert-base-uncased, fine-tuned	109,487,623
9	RoBERTa	Transformer	roberta-base, fine-tuned	124,651,015
10	BERT + RoBERTa ensemble	Transformer	Equal-weight soft voting of models 8 and 9	234,138,638

Training. BERT and RoBERTa were fine-tuned under one configuration (Table 3.3), and the checkpoint with the highest validation macro F1 was kept, which favours fairness to the rare classes. The classical models used balanced class weights. A class-weighted loss was also tested on RoBERTa as an ablation, with weights from 0.46 for Normal to 8.16 for Personality Disorder.

Table 3.3: Transformer training configuration

Setting	BERT and RoBERTa	Emotion-DistilRoBERTa
Optimiser	AdamW, learning rate 2×10^{-5} , weight decay 0.01	AdamW, learning rate 1×10^{-5} , weight decay 0.01
Batch size and epochs	32; 3 epochs	16; 5 epochs
Maximum input length	128 tokens	128 tokens
Precision	Mixed precision (fp16)	Mixed precision (fp16)
Checkpoint kept	Highest validation macro F1	Lowest validation loss
Hardware and seed	NVIDIA RTX 4060 laptop GPU (8 GB); seed 42	Same

Ensemble and explanation. The working model averages the class probabilities of BERT and RoBERTa with equal weight, and the class with the highest average probability becomes the prediction. The two models were pretrained differently, so they make partly different errors, which is when averaging helps [27].

The explanation is BERT’s attention from the classification token ([CLS]) to each word, averaged over all 12 layers and all 12 attention heads, and shown as a word heatmap. Because attention is not always a faithful explanation [32], [33], it was compared with LIME [30] and SHAP [31] on two example texts.

Risk scoring. Each class maps to a risk level and an emotional stress score for the fusion layer (Table 3.4). The distress score multiplies the probability of each class by its severity weight, adds the seven results and scales the total to a value from 0 to

100. A text that is clearly Suicidal therefore scores close to 100, and a clearly Normal text scores close to 0.

Table 3.4: Risk level, severity weight and emotional stress score of each class

Class	Risk level	Severity weight	Emotional stress score (0 to 4)
Normal	Low	0.00	0
Stress	Medium	0.50	1
Anxiety	Medium	0.60	2
Personality Disorder	High	0.70	3
Bipolar	High	0.75	3
Depression	High	0.80	3
Suicidal	Critical	1.00	4

Simulated lead-time test. The corpus has no time stamps, so a 16-week semester was simulated at Weeks 2, 4, 8 and 12. At each point, 50 Normal and 50 Depression passages were sampled and classified by BERT. The weeks remaining (14, 12, 8 and 4) show the size of the action window, not a measured outcome.

3.4 Measures and evaluation metrics

Accuracy, precision, recall and F1 were reported. Because the classes are imbalanced, both weighted F1, which weights each class by its size, and macro F1, which treats all classes equally, were used. Per-class results and a confusion matrix were reported for the working model.

3.5 Data analysis

95% confidence intervals were computed by bootstrap resampling of the test set with 2,000 resamples [64], and pairs of models were compared with McNemar’s test [65]. Logistic Regression and the SVM were also checked with five-fold cross-validation.

3.6 Ethics privacy security and responsible AI

No formal ethics approval was needed, because the study used only a public dataset with no personal identifiers and involved no participants. The following measures were applied:

- The component is decision support only and never diagnoses.

- Every alert is reviewed by a person, and Suicidal predictions are escalated as critical alerts.
- Every prediction comes with its explanation, so an advisor can question it.
- Use with students would need consent, pseudonymous IDs and role-based access (Section 8.4).
- The model may misread some writing styles, irony or sarcasm [40], so fairness should be checked before deployment.

3.7 Reproducibility and research artefacts

Code, notebooks and results are on GitHub (github.com/Sha20000/R26-IT-059-Burnout-Prediction), with large files in Git LFS. All experiments used the same split and seed (42), and every metric in this thesis comes from the saved results files. The main notebooks and scripts are listed in Table 3.5.

Table 3.5: Key notebooks and scripts

Notebook or script	Purpose
01_data_exploration	Loading, cleaning, duplicates and class distribution
02_baseline_models	TF-IDF baselines and five-fold cross-validation
03_lstm_model	BiLSTM model
04_bert_finetuning, 07_roberta_finetuning	Fine-tuning of BERT and RoBERTa, and the weighted-loss ablation
05_attention_visualization, 11_xai_comparison	Attention heatmaps and the LIME and SHAP comparison
06_vader_roberta_zeroshot	VADER and zero-shot baselines
08_mental_roberta_finetuning	Fine-tuning of Emotion-DistilRoBERTa
09_ensemble_model	Soft-voting ensemble, per-class results and confusion matrix
10_lead_time_analysis	Simulated lead-time test and the weekly examples
12_statistical_significance	Bootstrap confidence intervals and McNemar’s tests
serve_model.py, predict.py	Flask REST API (port 5004) and command-line prediction

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements came from the literature, supervisor and panel feedback, and team integration meetings. The main user is a student counsellor or academic advisor who is not a data scientist, so every output must be easy to read. The functional requirements are listed in Table 4.1 and the non-functional requirements in Table 4.2.

Table 4.1: Functional requirements

ID	Requirement
FR1	Accept a student's text through the API and the dashboard
FR2	Reject empty or very short text with a clear message
FR3	Classify the text into one of seven emotional states and return all seven probabilities
FR4	Give a risk level (Low, Medium, High or Critical) and a distress score from 0 to 100
FR5	Return word-level attention weights and show them as a heatmap
FR6	Show an advisor recommendation that matches the risk level
FR7	Show a student's results at Weeks 2, 4, 8 and 12
FR8	Export an emotional stress score (0 to 4) per student for the Meta-Integration layer
FR9	Report the service status and the measured model performance

Table 4.2: Non-functional requirements

ID	Requirement
NFR1	Accuracy: weighted F1 of at least 0.80 on the test set
NFR2	Explainability: every prediction carries word-level evidence
NFR3	Performance: well under one second per text
NFR4	Reliability: the live service uses the same models and settings as the evaluated ensemble
NFR5	Usability: outputs readable by non-technical staff
NFR6	Privacy: pseudonymous IDs, decision support only and human review of every alert
NFR7	Interoperability: JSON over HTTP and CSV export in the agreed team format

4.2 Component architecture within the complete solution

Figure 4.1 shows the R26-IT-059 system. Three components analyse academic records, student text and LMS behaviour, and the Meta-Integration layer fuses their outputs into one risk level and advisor action. Inside the emotional component (Figure 4.2), the text is cleaned and passed to the BERT and RoBERTa ensemble. Its seven class probabilities feed three outputs: the risk level and distress score, the attention explanation, and the emotional stress score for the fusion layer.

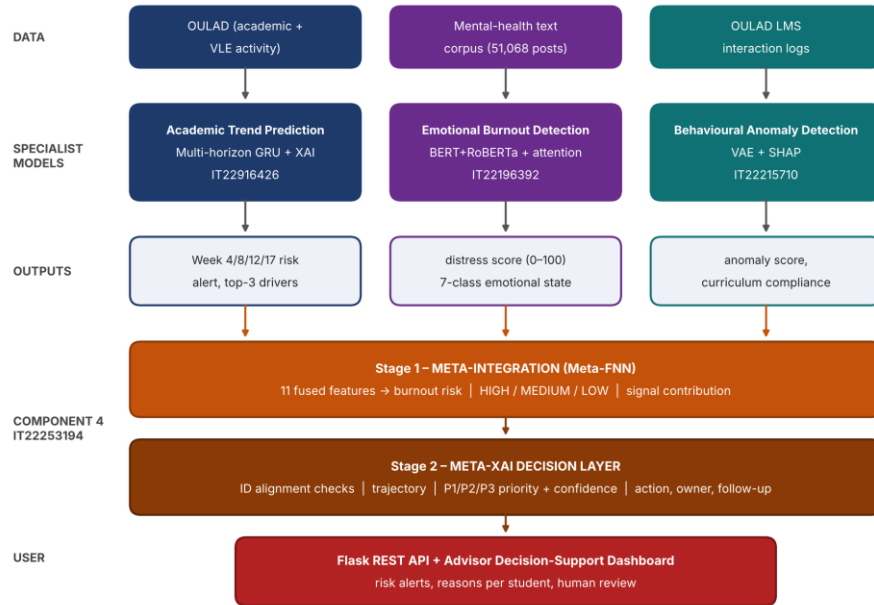


Figure 4.1: Overall architecture of the R26-IT-059 system; this thesis covers the Emotional Burnout Detection component (IT22196392). Reproduced from the team’s common report

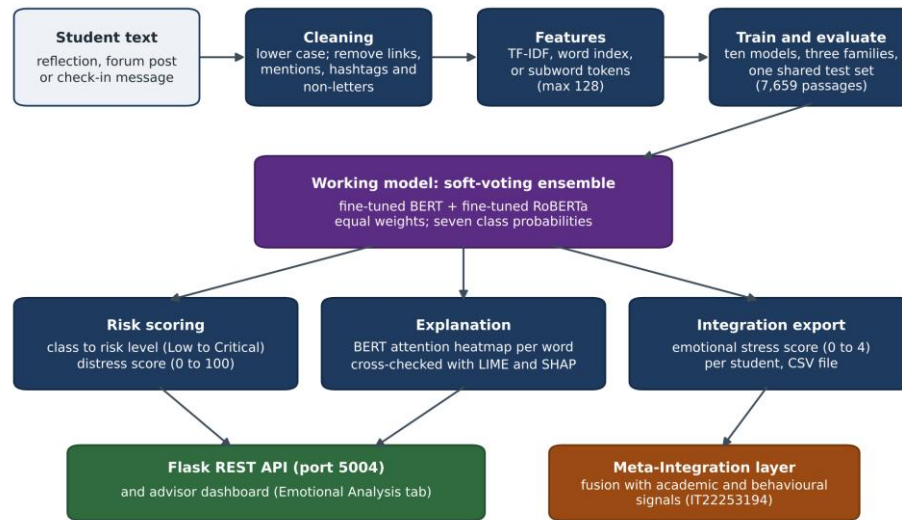


Figure 4.2: Internal design of the emotional component, from student text to the dashboard and the fusion layer

4.3 Design alternatives and decisions

The main design alternatives and the reasons for each choice are summarised in Table 4.3.

Table 4.3: Design decisions

Decision	Alternatives considered	Choice and reason
Model family	Lexicon, classical ML, BiLSTM, transformers	Fine-tuned transformers: highest scores (Table 6.1)
Single model or ensemble	RoBERTa alone	Equal-weight soft voting: same accuracy as RoBERTa, higher macro F1 (Table 6.3)
Model size	Large models (about 355 million parameters)	Base models: train on an 8 GB laptop GPU and run on an ordinary computer
Explanation method	LIME or SHAP only	Attention for live use, because it adds no time; LIME and SHAP used to check it
Class imbalance	Weighted loss, oversampling	Balanced weights and macro-F1 checkpoint selection; weighted loss tested but not adopted
Input length	64, 128 or 256 tokens	128 tokens: the median passage (61 words) fits and memory stays low
Service framework	FastAPI, Django	Flask: lightweight and the same as the team's other services
Integration style	Joint training of all components	Late fusion through an exported score, so each component can be replaced independently

4.4 Detailed design

For each text, BERT and RoBERTa run once and their probabilities are averaged (Section 3.3). The attention explanation comes from the same BERT run, so it adds no extra time, and the response also shows each model's own prediction. The predicted class sets the risk level and an advisor recommendation. For example, Stress shows "Recommend check-in with advisor", Depression shows "Immediate counseling recommended", and Suicidal is shown as a critical alert. The Emotional Analysis tab has three views: Analysis for a selected student, Weekly Progression across Weeks 2, 4, 8 and 12, and Quick Analysis for any typed text.

4.5 Standards and professional practice

Professional practices were followed so that the component stays maintainable, reproducible and trustworthy (Table 4.4).

Table 4.4: Standards and professional practices applied

Practice	How it was applied
Version control	Code, notebooks and results on GitHub, with Git LFS for model and data files
REST API design	JSON responses with standard status codes (200, 400, 500 and 503); separate health and model-information endpoints
Reproducibility	Fixed seed (42), one fixed stratified split and saved metric files
Fair evaluation	One shared test set, confidence intervals, significance tests and an ablation reported even though it did not help
Structured testing	Test cases with input, expected result and status (Section 5.4)
Responsible AI	Decision support only, human review of every alert and an explanation with every prediction (Section 3.6)
Referencing	All sources cited in the IEEE style

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The component was built in Python and trained on a laptop with an NVIDIA RTX 4060 GPU (8 GB). The technologies used are listed in Table 5.1.

Table 5.1: Technology stack

Layer	Technology	Purpose
Language	Python 3; JavaScript	Models and API; dashboard
Deep learning	PyTorch; Hugging Face Transformers [66], [67]	BiLSTM and transformer models
Classical ML	scikit-learn [68]	TF-IDF baselines, cross-validation and metrics
Baselines	vaderSentiment; facebook/bart-large-mnli	Untrained baselines
Explainability	Transformer attention; LIME; SHAP	Word-level explanations and their check
Data and charts	pandas, NumPy, Matplotlib, Seaborn	Data handling and figures
Service	Flask with Flask-CORS	REST API on port 5004
Frontend	HTML, CSS, JavaScript; Next.js and React	Team dashboard and standalone interface
Tools	Jupyter, VS Code, Git, GitHub, Git LFS	Development and version control

5.2 Component implementation

The models were trained in the notebooks listed in Table 3.5 with the Hugging Face Trainer, and every result was saved as a file. The trained ensemble is served by a Flask REST API (`serve_model.py`, port 5004), which loads both models once at start-up and keeps them in memory. Listing 5.1 shows the core of the prediction function.

**Listing 5.1: Ensemble prediction, distress score and attention explanation
(serve_model.py, shortened)**

```
def analyze_text(text):
    inputs = TOKENIZER(text, return_tensors="pt", truncation=True, max_length=128)
    roberta_inputs = ROBERTA_TOKENIZER(text, return_tensors="pt",
                                       truncation=True, max_length=128)

    with torch.no_grad():
        outputs = MODEL(**inputs)          # BERT, attentions on
        roberta_outputs = ROBERTA_MODEL(**roberta_inputs)

    bert_probs = torch.softmax(outputs.logits, dim=1)[0]
    roberta_probs = torch.softmax(roberta_outputs.logits, dim=1)[0]
    probs = BERT_WEIGHT * bert_probs + ROBERTA_WEIGHT * roberta_probs # 0.5 / 0.5

    predicted_label = LABELS[int(probs.argmax().item())]
    probabilities = {LABELS[i]: round(float(p), 4) for i, p in enumerate(probs)}
    stress_score = round(sum(probabilities.get(c, 0) * w
                           for c, w in STRESS_WEIGHTS.items()) * 100, 1)

    # [CLS] row of the attention, averaged over 12 layers and 12 heads
    all_attentions = torch.stack(outputs.attentions)
    avg_attention = all_attentions.mean(dim=0).mean(dim=1).squeeze()
    cls_attention = avg_attention[0].numpy()
    tokens = TOKENIZER.convert_ids_to_tokens(inputs["input_ids"][0])
    clean_tokens, clean_weights = tokens[1:-1], cls_attention[1:-1]
    clean_weights = clean_weights / clean_weights.sum()
    ...
```

Both models read the same text, and their probabilities are averaged with equal weights. The distress score is computed from all seven probabilities, and the attention comes from the same BERT run. Listing 5.2 shows the analysis endpoint, which checks the input before calling the model, and Table 5.2 lists all the endpoints.

Listing 5.2: Analysis endpoint with input validation (serve_model.py)

```
@app.route("/api/analyze", methods=["POST", "OPTIONS"])
def analyze():
    if request.method == "OPTIONS":
        return ("", 200)
    if MODEL is None:
        return jsonify({"error": "Model not loaded."}), 503
    data = request.get_json(silent=True) or {}
    text = (data.get("text") or "").strip()
    if not text or len(text.split()) < 3:
        return jsonify({"error": "Please enter at least a few words"}), 400
    try:
        return jsonify(analyze_text(text))
    except Exception as e:
        return jsonify({"error": "Model inference failed.", "detail": str(e)}), 500
```

Table 5.2: REST API endpoints of the emotional component

Endpoint	Purpose
POST /api/analyze	Analyse one text: prediction, confidence, risk level, distress score, all seven probabilities, word attention and each model's prediction
GET /api/health	Service status, the classes and the ensemble settings
GET /api/model-info	Measured test performance, read from the saved results file

The first version of the service, shown at the progress presentations, served BERT alone. The final version loads both models, so the live service matches the evaluated ensemble.

The dashboard shows each result in the Emotional Analysis tab. For an exam-stress text (Figure 5.1), the ensemble predicts Stress with 59.4% confidence, a distress score of 40.6 and a Medium risk level, and the heatmap marks “stressed” and “cannot”. For a text about exhaustion and hopelessness (Figure 5.2), it predicts Depression (56.6%) with Suicidal close behind (41.8%), and the distress score of 87.4 reflects both. Figure 5.3 shows the Weekly Progression view for a demonstration student, with a risk line across the four checkpoints.

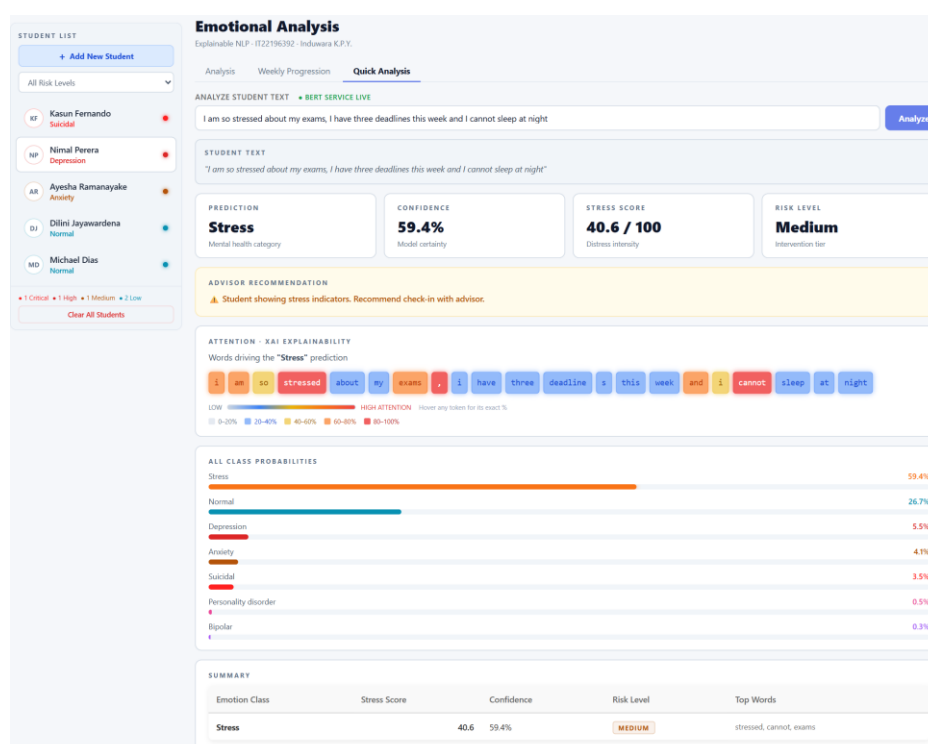


Figure 5.1: Quick Analysis view for an exam-stress text: prediction, confidence, distress score, risk level, recommendation, attention heatmap and class probabilities

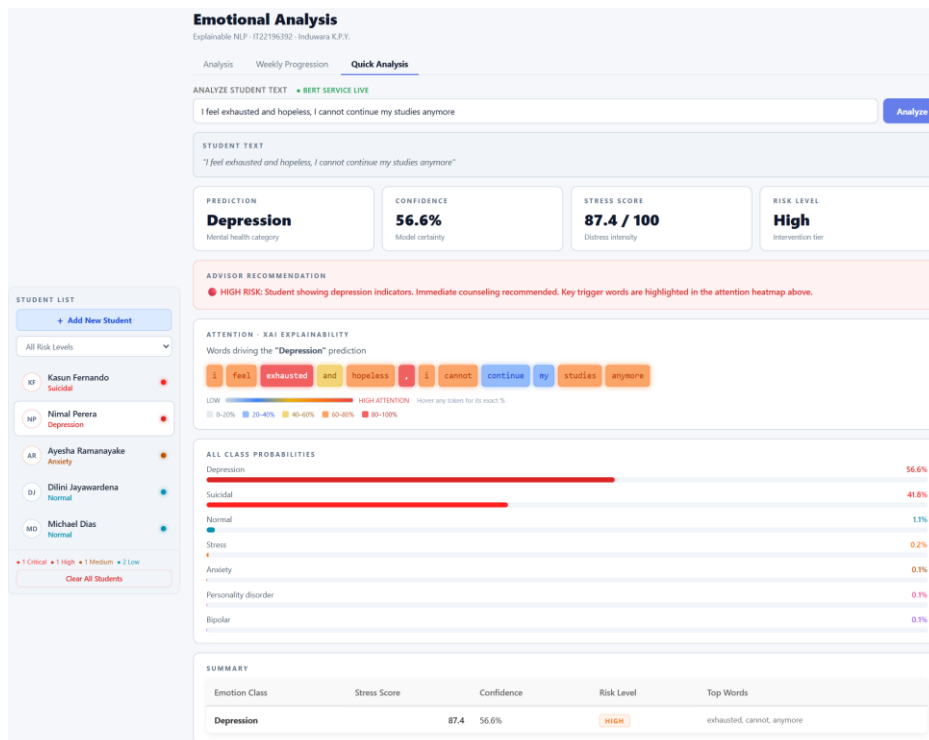


Figure 5.2: Quick Analysis view for a text expressing exhaustion and hopelessness, with Depression and Suicidal as the two most likely classes

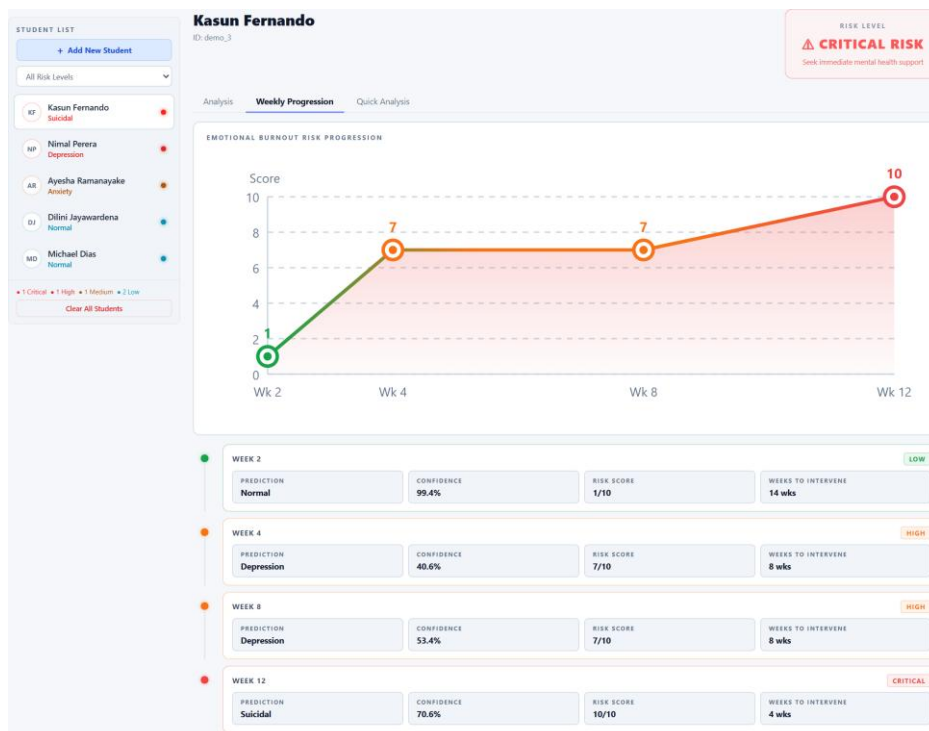


Figure 5.3: Weekly Progression view for a demonstration student, with the risk line and the result of each checkpoint

5.3 Integration with the other components

The component connects to the team system in two ways (Figure 5.4). The dashboard calls the emotional API directly, and the component exports an emotional stress score file in the agreed CSV format (Table 5.3). The four team services run side by side on ports 5001 to 5004, with the emotional API on port 5004.

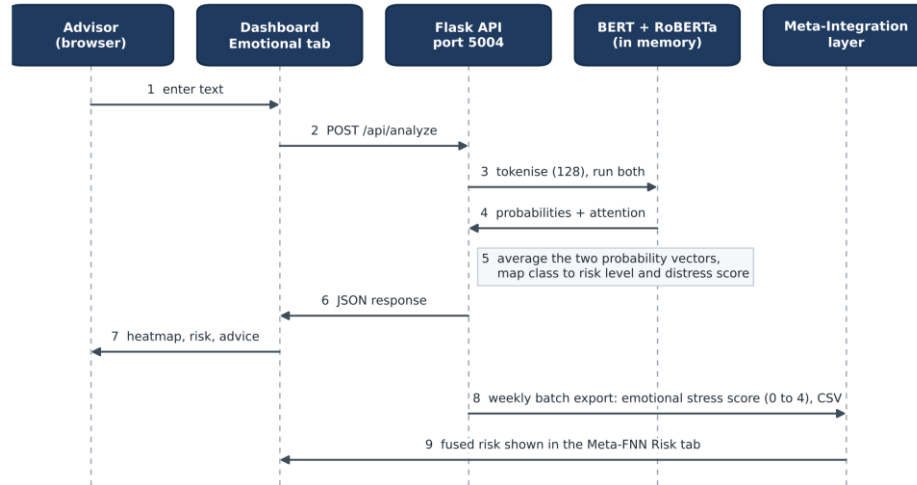


Figure 5.4: Data flow between the advisor, the dashboard, the emotional API, the models and the Meta-Integration layer

Table 5.3: Data contract of the emotional score file passed to the Meta-Integration layer

Field	Meaning
student_id	Pseudonymous student identifier (for example STU1000)
text_sample	Short extract of the analysed text
predicted_label	One of the seven classes
confidence	Probability of the predicted class
emotional_stress_score	Integer from 0 (Normal) to 4 (Suicidal), as in Table 3.4
risk_level	Risk level of the predicted class

The Meta-Integration layer uses `emotional_stress_score` as one of its eleven input features, together with five academic and five behavioural features. Figure 5.5 shows the emotional score next to the other signals in the Meta-FNN Risk tab. The 500 exported records were produced from the public corpus with pseudonymous IDs, because no public dataset links student writing to the OULAD records used by the other components [69].

Test Students	PP2 Concurrent	Model Metrics	Meta-FNN Risk	Emotional Analysis	Academic Risk	Behavioral Anomaly	SLIT Synthetic			
FILTER BY ALERT LEVEL										
All Students										
Student ID	Academic Risk (GRU)	Week 4	Week 8	Week 12	Week 17	Behavioral Risk (VAE)	Emotional Risk (NLP)	Final Burnout Risk	Weighted Score	Alert Level
OULAD_TEST_0000	25.6%	50.2%	19.6%	37.0%	25.6%	22.9%	1.3 / 4	23.5%	27.0%	LOW LOW
OULAD_TEST_0001	3.3%	23.9%	15.1%	7.8%	3.3%	0.9%	0.0 / 4	6.8%	2.0%	LOW LOW
OULAD_TEST_0002	6.4%	47.4%	35.3%	23.7%	6.4%	11.0%	0.6 / 4	9.4%	10.7%	LOW LOW
OULAD_TEST_0003	39.2%	71.2%	53.7%	48.0%	39.2%	42.7%	2.5 / 4	61.1%	47.1%	MEDIUM MEDIUM
OULAD_TEST_0004	10.8%	33.9%	16.6%	9.8%	10.8%	5.2%	0.3 / 4	9.4%	8.1%	LOW LOW
OULAD_TEST_0005	25.1%	67.4%	50.7%	23.7%	25.1%	15.2%	0.9 / 4	32.9%	21.1%	LOW LOW
OULAD_TEST_0006	22.2%	47.7%	47.3%	38.9%	22.2%	31.3%	1.8 / 4	29.2%	32.0%	LOW LOW
OULAD_TEST_0007	99.2%	44.2%	74.4%	83.6%	99.2%	77.1%	4.0 / 4	90.2%	92.8%	HIGH HIGH
OULAD_TEST_0008	39.8%	77.8%	80.5%	71.9%	39.8%	23.1%	1.3 / 4	48.0%	32.7%	MEDIUM LOW
OULAD_TEST_0009	41.4%	29.7%	18.7%	25.4%	41.4%	34.4%	2.0 / 4	43.6%	41.8%	MEDIUM MEDIUM
OULAD_TEST_0010	86.5%	52.5%	34.0%	39.6%	86.5%	55.9%	3.2 / 4	80.2%	75.2%	HIGH HIGH
OULAD_TEST_0011	99.9%	84.0%	94.1%	98.0%	99.9%	65.3%	3.7 / 4	93.4%	87.4%	HIGH HIGH

Figure 5.5: Meta-FNN Risk tab of the team dashboard, showing the emotional score (out of 4) next to the academic and behavioural signals

5.4 Verification and software testing

The component was checked with unit tests on the data and model outputs, integration tests on the API and the export file, and user acceptance tests on the dashboard (Table 5.4).

Table 5.4: Test cases

ID	Type	Test	Input	Expected and actual result	Status
TC-01	Unit	Corpus size	cleaned_data.csv	51,068 rows; 13 empty after cleaning; 51,055 used	Pass
TC-02	Unit	Labels	Label column	Exactly seven classes with the counts in Figure 3.1	Pass
TC-03	Unit	Split sizes	Stratified split, seed 42	35,738 / 7,658 / 7,659 passages	Pass
TC-04	Unit	Test support	Test labels	Normal 2,405; Depression 2,263; Suicidal 1,595; Anxiety 543; Bipolar 375; Stress 344; Personality Disorder 134	Pass
TC-05	Unit	Reproducibility	Saved models, test set	Ensemble accuracy 82.32% and weighted F1 0.8232 reproduced	Pass
TC-06	Unit	Score mapping	500 exported records	Every class maps to the stress score in Table 3.4	Pass
TC-07	Integration	Health check	GET /api/health	HTTP 200; both models loaded; weights 0.5 and 0.5	Pass
TC-08	Integration	Model information	GET /api/model-info	Accuracy 82.32%, macro F1 78.93%, 7,659 test passages	Pass
TC-09	Integration	Stress text	POST /api/analyze	Stress (59.4%), Medium, distress 40.6; both models predict Stress	Pass
TC-10	Integration	Depression text	POST /api/analyze	Depression (56.6%), High, distress 87.4; Suicidal second (41.8%)	Pass
TC-11	Integration	Export file	emotional_stress_scores.csv	500 records, IDs STU1000 to STU1499, no missing values	Pass

ID	Type	Test	Input	Expected and actual result	Status
TC-12	Integration	Fusion input	Meta-layer configuration	emotional_stress_score is one of the eleven fusion features	Pass
TC-13	UAT	Quick Analysis	Dashboard, typed text	Prediction, risk, recommendation, heatmap and probabilities shown (Figure 5.1)	Pass
TC-14	UAT	Weekly view	Dashboard, demonstration student	Week-by-week results and risk line shown (Figure 5.3)	Pass

All fourteen tests passed. Testing also found two problems, which were fixed: the first service served BERT under the ensemble’s name, and the dashboard showed a “Week 18” checkpoint, which was corrected to Week 12.

5.5 Deployment and operation

The component runs as a local prototype. The API keeps both models in memory, the dashboard runs in a web browser, and the scores for the fusion layer are produced in a batch run, so the dashboard keeps working while a model is retrained. The operational workflow is summarised in Table 5.5.

Table 5.5: Operational workflow

Step	Action	Output
1	Start the emotional API (python <code>serve_model.py</code>)	Both models loaded; service on port 5004
2	Open the team dashboard in a browser	Emotional Analysis tab connected to the API
3	Analyse a text or open a student’s weekly view	Prediction, risk, distress score, recommendation and heatmap
4	Run the batch scoring of student texts	<code>emotional_stress_scores.csv</code>
5	Run the Meta-Integration layer	Fused risk and alert level in the Meta-FNN Risk tab

6 RESULTS AND VALIDATION

6.1 Research results

All results come from the saved results files and were measured on the same test set of 7,659 passages unless stated otherwise.

Model comparison (RQ1). Table 6.1 and Figure 6.1 show three clear groups. All four fine-tuned transformers reached a weighted F1 above 0.80, the classical models and the BiLSTM reached between 0.699 and 0.766, and the two untrained models reached 0.344 or less. Five-fold cross-validation gave almost the same results for Logistic Regression (75.88% accuracy) and the SVM (74.83%), so the baselines do not depend on one split. Emotion pretraining did not help: Emotion-DistilRoBERTa scored below BERT and RoBERTa, because basic emotions such as sadness or fear do not map directly onto clinical categories.

Table 6.1: Accuracy, weighted F1 and macro F1 of all ten models on the test set

#	Model	Group	Accuracy	Weighted F1	Macro F1
1	VADER	Untrained	25.02%	0.2279	0.1251
2	Zero-shot (BART-large-MNLI)*	Untrained	32.80%	0.3439	0.3071
3	Random Forest	Classical ML	71.87%	0.6994	–
4	Support Vector Machine	Classical ML	75.49%	0.7498	–
5	Logistic Regression	Classical ML	76.69%	0.7595	–
6	BiLSTM	Deep learning	76.84%	0.7662	0.69
7	Emotion-DistilRoBERTa	Transformer	80.00%	0.8013	0.7446
8	BERT	Transformer	81.39%	0.8143	0.7767
9	RoBERTa	Transformer	82.30%	0.8231	0.7826
10	BERT + RoBERTa ensemble	Transformer	82.32%	0.8232	0.7893

* Scored on a stratified subset of 500 test passages; all other models used all 7,659. A dash means the value was not recorded for that model.

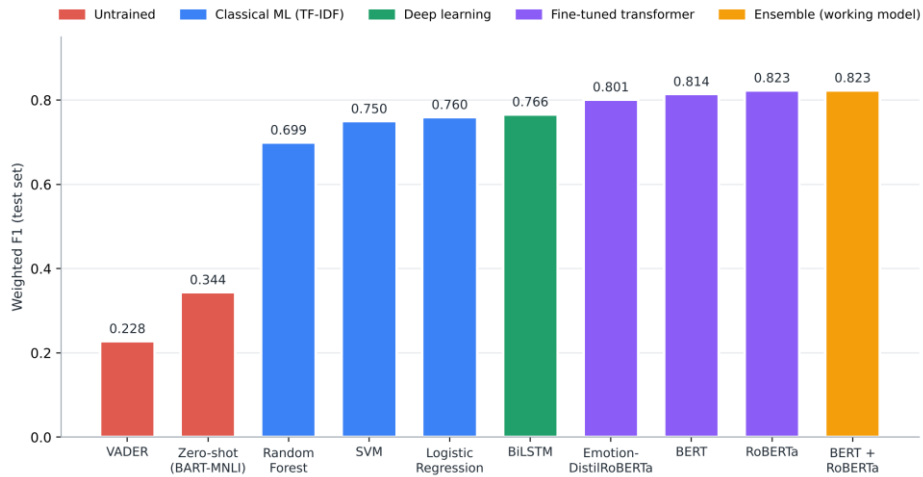


Figure 6.1: Weighted F1 of all ten models on the test set, coloured by model group

Working model. The ensemble had the highest value on all three measures (82.32% accuracy, weighted F1 0.823 and macro F1 0.789). Table 6.2 and Figure 6.2 show its results for each class. Normal was almost always correct (F1 0.955), and the smallest class, Personality Disorder, was the hardest (F1 0.651). The main confusion was between Depression and Suicidal: 21.6% of Depression passages were predicted as Suicidal and 21.3% of Suicidal passages as Depression. These classes share much of the same language, and both lead to a High or Critical alert and human review.

Table 6.2: Per-class results of the BERT + RoBERTa ensemble on the test set

Class	Precision	Recall	F1	Test passages	Risk level
Normal	0.952	0.958	0.955	2,405	Low
Anxiety	0.850	0.888	0.869	543	Medium
Bipolar	0.864	0.816	0.840	375	High
Depression	0.788	0.741	0.764	2,263	High
Suicidal	0.702	0.752	0.726	1,595	Critical
Stress	0.700	0.744	0.721	344	Medium
Personality Disorder	0.695	0.612	0.651	134	High
Macro average	—	—	0.789	7,659	—

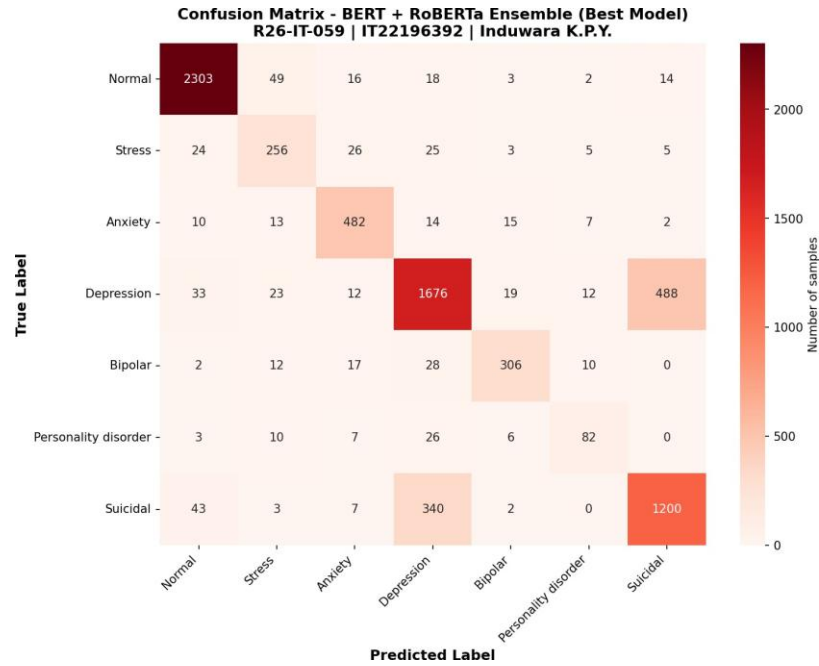


Figure 6.2: Confusion matrix of the BERT + RoBERTa ensemble on the test set

Statistical tests (RQ2). The ensemble was significantly more accurate than BERT and Emotion-DistilRoBERTa, but its accuracy was statistically the same as RoBERTa's ($p = 0.96$), and their confidence intervals overlap almost completely (Table 6.3, Table 6.4). The gain of the ensemble is in class balance: it has the highest macro F1, because averaging helps most on the rarer classes.

Table 6.3: Accuracy and macro F1 with 95% bootstrap confidence intervals ($n = 7,659$, 2,000 resamples)

Model	Accuracy (95% CI)	Macro F1 (95% CI)
Emotion-DistilRoBERTa	80.00% (79.06 to 80.87)	0.745 (0.729 to 0.760)
BERT	81.39% (80.51 to 82.24)	0.777 (0.762 to 0.791)
RoBERTa	82.30% (81.42 to 83.13)	0.783 (0.767 to 0.797)
BERT + RoBERTa ensemble	82.32% (81.46 to 83.17)	0.789 (0.775 to 0.803)

Table 6.4: McNemar's paired tests on the test set

Comparison	Only A correct	Only B correct	p value	Result
Ensemble (A) vs BERT (B)	210	139	0.0002	Significant
Ensemble (A) vs Emotion-DistilRoBERTa (B)	427	249	< 0.001	Significant
BERT (A) vs RoBERTa (B)	298	367	0.008	Significant (RoBERTa better)
Ensemble (A) vs RoBERTa (B)	160	158	0.96	Not significant

Class-weighted loss. The weighted loss raised recall on Suicidal, Stress and Personality Disorder, but it also caused more false alarms and lowered Depression recall, accuracy (82.3% to 80.4%) and macro F1 (0.783 to 0.766). The plain model was therefore kept (Table 6.5).

Table 6.5: Plain and class-weighted RoBERTa on the test set

Measure	Plain RoBERTa (kept)	Class-weighted RoBERTa
Accuracy	82.30%	80.35%
Macro F1	0.783	0.766
Recall: Suicidal	0.75	0.81
Recall: Stress	0.75	0.83
Recall: Personality Disorder	0.55	0.61
Recall: Depression	0.75	0.67
Precision: Stress	0.68	0.59

Explanations (RQ3). For a text expressing depression, the highest attention went to emotionally loaded words such as “exhausted”, “cannot” and “anymore” (Figure 6.3). LIME and SHAP picked out the same strong words: for the depression text all three methods ranked “anymore” and “hopeless” highly, and for the stress text “stressed” was the top word in all three (Figure 6.4). Raw attention also gives some weight to small words and punctuation, so the dashboard removes them before listing the top words.

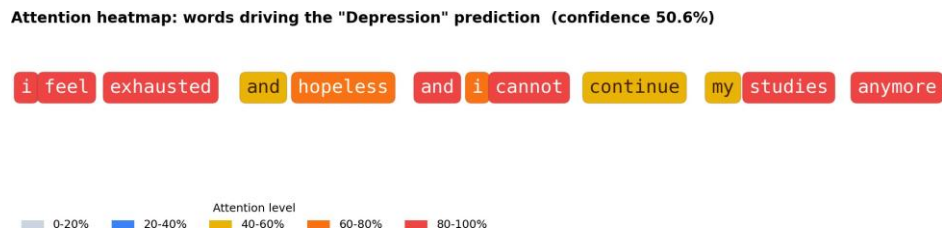


Figure 6.3: Word-level attention heatmap of the fine-tuned BERT model for a text expressing depression

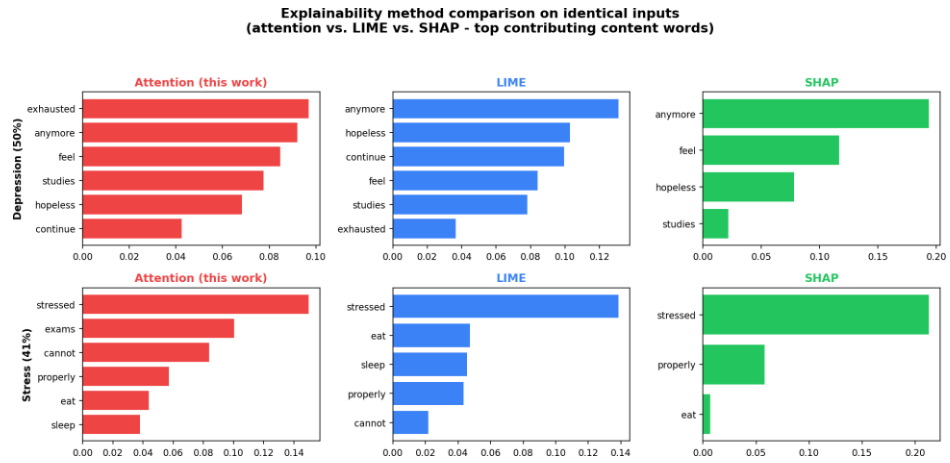


Figure 6.4: Top words found by attention (this work), LIME and SHAP for the same two inputs

Simulated lead time (RQ4). At every simulated checkpoint, BERT separated Normal from Depression texts with 89% to 91% accuracy (Table 6.6, Figure 6.5). Many of the sampled passages were also training passages, so these scores are optimistic. Four example sentences written by the author moved from Normal at Week 2 to Stress, Depression and Suicidal at Week 12, with the risk level rising at each step (Figure 6.6).

Table 6.6: Simulated lead-time test (100 passages per checkpoint: 50 Normal and 50 Depression)

Checkpoint	Accuracy	Weighted F1	Weeks remaining
Week 2	91%	0.9505	14 weeks
Week 4	89%	0.9344	12 weeks
Week 8	91%	0.9515	8 weeks
Week 12	89%	0.9382	4 weeks

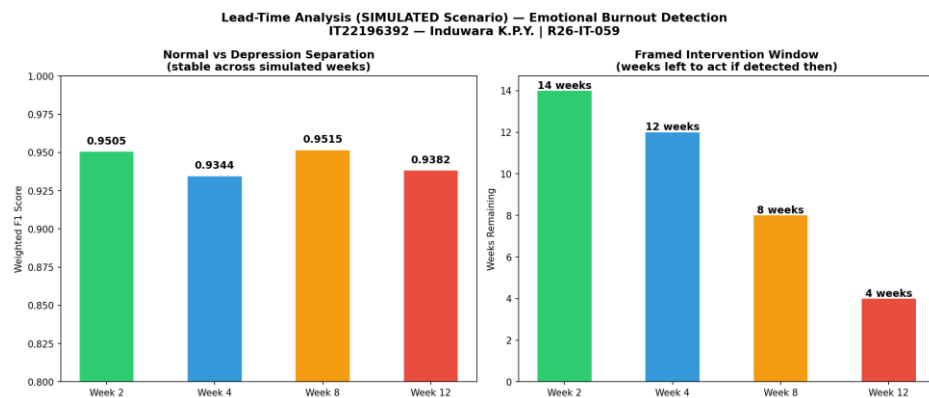


Figure 6.5: Simulated lead-time test: separation of Normal and Depression texts at four checkpoints, and the intervention window

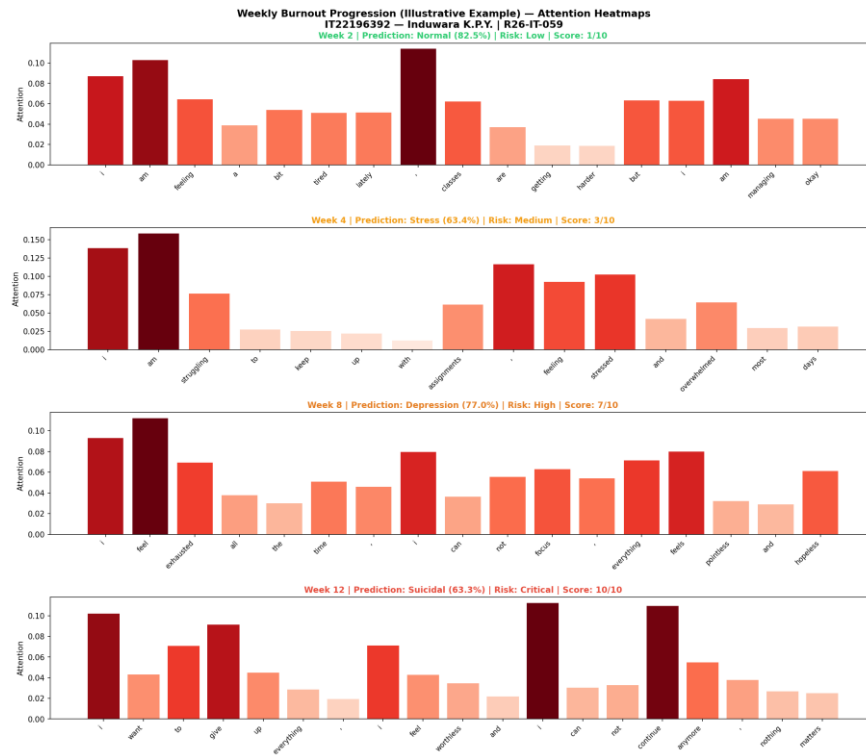


Figure 6.6: Weekly example: the prediction, risk level and attention for four author-written sentences

6.2 Product and technical results

The component met its functional requirements, and all fourteen test cases passed (Table 5.4). The health endpoint confirmed that both models were loaded, and the model-information endpoint returned the measured test results. Table 6.7 shows the live results for the two test texts; BERT and RoBERTa agreed on both, with RoBERTa more confident. During evaluation on the RTX 4060, BERT classified a passage in about 23 milliseconds and the ensemble in about 50 milliseconds, which meets NFR3.

Table 6.7: Live API results for two test texts

Text	Prediction (ensemble)	BERT alone	RoBERTa alone	Distress score	Risk
“I am so stressed about my exams, I have three deadlines this week and I cannot sleep at night”	Stress (59.4%)	Stress (43.4%)	Stress (75.4%)	40.6	Medium
“I feel exhausted and hopeless, I cannot continue my studies anymore”	Depression (56.6%)	Depression (50.8%)	Depression (62.4%)	87.4	High

6.3 User or stakeholder validation

The dashboard and the API were demonstrated to the supervisors and the evaluation panel at the progress presentations. Their feedback led to clearer framing of the lead-time result as a simulation and of the dataset as a proxy, the removal of novelty claims, and more weight on the explanation of each prediction. A formal study with counsellors is the next step (Section 9.4).

6.4 Integration results

The Meta-Integration layer used the emotional score as one of its eleven input features. The fused results reported in the team's common report are shown in Table 6.8. A permutation analysis attributed about 15% of the fused prediction to the emotional signal. Because the emotional scores and the academic records come from different datasets [69], these results show that the full pipeline works, from student text to a fused alert.

Table 6.8: Results of the fusion prototype (Meta-Integration FNN, threshold 0.45), as reported in the common report

Measure	Value
F1	0.6836
AUC	0.8576
Accuracy	77.23%
Precision / recall	0.6065 / 0.7833
Share of the fused prediction: academic / behavioural / emotional	55% / 30% / 15%
Alert distribution: HIGH / MEDIUM / LOW	about 29% / 15% / 56%

6.5 Summary of findings

- RQ1: Fine-tuned transformers scored higher than the other model groups, with a weighted F1 above 0.80.
- RQ2: The ensemble reached 82.32% accuracy and a macro F1 of 0.789. It was level with RoBERTa in accuracy but more balanced across the rare classes.
- RQ3: Attention highlighted meaningful words and agreed with LIME and SHAP on the examples checked.
- RQ4: Distress language was separated from ordinary language at every simulated checkpoint, and the API, dashboard and fusion export work from end to end.

7 DISCUSSION

7.1 Interpretation of findings

The results form a clear ladder. Models with no training on the task were close to guessing, word-statistics models reached about 75% accuracy, and fine-tuned transformers added about six more points, because they read each word in the context of the whole text. The ensemble should be described as giving the most balanced result rather than clearly higher accuracy than RoBERTa. The weakest classes were also the smallest, and loss weighting only moved errors around, so more data is the way to improve them. Attention gives a quick, readable view of the evidence, which an advisor should treat as a pointer to the important words rather than as proof.

7.2 Comparison with literature baselines and existing solutions

Table 7.1 compares this work with the closest studies. The highest scores in the literature come from binary tasks [6], [10], which are much easier than separating seven overlapping states. The closest study is by Khan et al., who used the same seven labels and reported 97% accuracy with RoBERTa-Large [12]. That model is about three times larger than the base models used here, and the two studies also differ in cleaning and splitting; for example, this study removed 1,612 duplicate texts before splitting. What this work adds is an explanation for every prediction, checked against LIME and SHAP, together with statistical tests of the model differences.

Table 7.1: This work compared with related studies

Study	Data	Classes	Model	Reported result	Explanation of each prediction
Beegam and Baalaji [6]	Kaggle depression text, 7,489 posts	2	BERT with a deep neural network	Accuracy 97%	No (dataset word clouds)
Merhbene et al. [10]	Reddit posts	2	Logistic regression ensembles	Balanced accuracy 0.93, F1 0.43	No
Mazurets et al. [11]	Tweets of healthcare workers	2	BERT with LDA topics	Accuracy 0.83, F1 0.87	Topic level only
Zhuang et al. [40]	Postgraduate social-media posts	Stress profile	BERT, LDA and BiLSTM-CRF	Accuracy 92.55%, F1 92.18%	No
Khan et al. [12]	Kaggle mental-health corpus	7	RoBERTa-Large	Accuracy 97%, F1 0.94	No
This thesis	Kaggle mental-health corpus	7	BERT and RoBERTa ensemble	Accuracy 82.32%, macro F1 0.789	Yes, word heatmap checked with LIME and SHAP

Among existing tools, LMS analytics and student-success platforms give little or no reason for a flag [61], [55], and school safety tools work by monitoring school devices [62], [63]. Workplace survey tools summarise comments by topic [59]. This component is designed for text that people choose to share, with a graded emotional state and readable evidence for each alert (Table 8.2).

7.3 Research contribution

- A comparison of ten models on one test set for seven-class mental-health text, with confidence intervals and McNemar’s tests.
- A seven-class ensemble in which every prediction comes with a word-level explanation, checked against LIME and SHAP.
- A simulated lead-time test showing that distress language can be picked up at any point in a semester.

7.4 Product contribution

A working REST API and Emotional Analysis dashboard that give the emotional state, risk level, distress score and word evidence for any text, a score export used by the team’s fusion layer, and a commercialisation plan for universities and workplaces (Chapter 8).

7.5 Limitations and threats to validity

- The labels come from the source datasets [7] and the texts are public social-media posts in English, so the component gives a signal for review and should be checked on consented student text.
- The split was made at passage level, and the lead-time samples included training passages, so some scores are slightly optimistic.
- The zero-shot baseline used 500 passages, and the confidence intervals do not cover different training seeds.
- Attention is not always a faithful explanation [32], and the LIME and SHAP check covered two examples.
- The first service read raw text, while the models were trained on cleaned text.
- The emotional scores and the academic records come from different datasets, so the fused results show that the pipeline works rather than how much each signal adds.

Steps already taken. A second version of the component was built to address some of these points. It is reported separately from the main results:

- The service applies the same cleaning as training, and both models read up to 256 tokens (the first 128 and the last 126 of long posts).
- The public corpus was arranged into a simulated weekly cohort of 13,171 pseudonymous students with texts at Weeks 2, 4, 8 and 12, split by student, so the full weekly pipeline could be tested.
- The retrained ensemble (RoBERTa 0.55 and BERT 0.45) reached 85.25% accuracy and a macro F1 of 0.843 on 7,887 held-out texts from 1,974 students. It uses a different split, so it is not compared directly with the main results.

7.6 Implications

Student writing can add an emotional signal that grades and LMS activity do not show, if students opt in and every alert is reviewed by a trained person, with critical cases passed to mental-health staff. For researchers, gains from ensembles and loss weighting should be tested statistically, and explanations should be checked against independent methods.

8 COMMERCIALISATION AND SUSTAINABILITY

This chapter looks at the component as a product. It was built for university students, but the engine inside it is more general. It reads a short piece of text, tells which emotional state the text shows and how serious it is, and shows the words behind the result. Burnout is not only a student problem: the World Health Organization describes burn-out as the result of chronic workplace stress that has not been successfully managed [56]. The same engine can therefore help in any place where people write about how they feel and where someone is responsible for their well-being. The plan is to start with universities and then grow to workplaces, hospitals and app partners.

8.1 Value proposition and target users

8.1.1 The product and its value

The product has three parts, and a first version of each already exists:

- **The analysis engine.** A REST API that takes a text and returns one of seven emotional states, a risk level, a distress score from 0 to 100, the probability of each state and the words behind the result.
- **The advisor dashboard.** Quick Analysis for any typed text, a weekly view for each person, and a recommended next step for the advisor.
- **Connectors.** Score exports in CSV and JSON, the link to the team's fusion layer, and planned plugins for Moodle and staff survey tools.

Value proposition. For counsellors, advisors and well-being officers who look after a large group of people, the engine turns the short texts that people choose to share into an early and explained warning of emotional strain. Instead of a yes-or-no flag, it gives one of seven emotional states, a risk level and a distress score, and it shows the words behind every alert. A trained person then decides what to do. Suicidal predictions are raised as critical alerts, so that the person can be referred at once, for example to the national mental health helpline 1926 in Sri Lanka [70].

8.1.2 Target markets

The engine can serve several markets. Table 8.1 lists them in the planned order of entry, starting with higher education, where the research evidence was built.

Table 8.1: Target markets for the emotional burnout engine

Market	Who uses it	Text analysed (always opt-in)	Main value	Order
Universities and private institutes	Student counsellors and academic advisors	Weekly reflections and check-in forms	Finds students in distress early, next to their academic risk	1 (Year 1)
Companies, starting with IT and BPO firms	Well-being officers and workplace counsellors	Well-being check-ins and staff survey comments	Team-level burnout trends and confidential support	2 (Year 2)
Mental-health and journaling apps	App developers, through the API	Journal entries and chat messages, with the user's consent	Explained risk levels and a safety alert inside the app	3 (Year 2 to 3)
Hospitals and care homes	Occupational health and staff welfare units	Short reflections after shifts	Early signs of burnout among nurses, doctors and other staff	4 (Year 3)
Online learning platforms	Course support teams	Forum posts and course feedback	Notifies learners who are struggling emotionally	5 (Year 3)

Higher education comes first because the research was built for it, the team platform already serves universities, and a pilot can start at SLIIT. Companies come second, because they need only a new connector, a team-level dashboard and stronger privacy rules, and many IT and BPO companies already run staff surveys. Apps, hospitals and online learning follow once the engine has been checked on consented data from the first pilots.

8.1.3 User personas

Persona 1: student counsellor. Mr. Chamara Silva (an illustrative persona) is a student counsellor at a private higher education institute with about 4,000 students and two counsellors. Students write a short weekly reflection in the learning management system, but he cannot read them all. Most students who are struggling reach him only after they miss assessments. He needs a short weekly list of students whose writing shows distress, how serious each case is, and the words behind each flag, so that he can decide whom to contact first.

Persona 2: workplace well-being lead. Ms. Nadeesha Perera (an illustrative persona) leads employee well-being at an IT company with about 800 staff. Every month the company runs a short survey with one open question about workload and

stress. She reads hundreds of comments by hand, so she often learns about burnout in a team only when people resign. She needs team-level trends with the reasons behind them, while every person stays anonymous to managers.

8.2 Market and competitive position

8.2.1 Market need

Education. In a survey of first-year university students in eight countries, about one in three reported a mental disorder in the previous year [4], and burnout is linked to lower academic achievement [3]. In Sri Lanka, higher education is offered by state universities and a growing number of private degree-awarding institutes, many of which use an LMS such as Moodle. Their counselling units are small compared with the number of students, so a tool that helps counsellors focus their time has clear value.

Workplaces. The World Health Organization and the International Labour Organization estimate that about 12 billion working days are lost every year to depression and anxiety, at a cost of nearly one trillion US dollars, and that 15% of working-age adults had a mental disorder in 2019 [71]. Because burn-out is defined as a workplace syndrome [56], employers have a direct reason to notice it early.

Healthcare. Health workers are under strong pressure. Nurses' tweets showed more burnout and stress language than those of construction workers [57], and talk about work on the hospital unit was linked with anxiety and depression in the therapy messages of healthcare workers [58].

At the same time, any product that reads what people write must earn their trust. People worry about privacy and surveillance in schools [63] and at work, so the design of the product, and not only its accuracy, decides whether people will use it.

8.2.2 Competitors and alternatives

Table 8.2 compares the engine with the main types of existing tools. Institutional predictive tools such as OU Analyse [55] and work-pattern tools such as Microsoft Viva Insights [60] are similar to the LMS analytics and survey columns: they look at activity or habits, not at the emotional content of text.

Table 8.2: Comparison with existing solutions

Feature	LMS analytics (Moodle) [61]	Student-success platforms (EAB Navigate, Civitas Learning)	School safety monitoring (GoGuardian Beacon, Gaggle) [62], [63]	Staff survey tools (Workday Peakon) [59]	This engine
Main data	LMS activity	Academic and enrolment data	Online activity on school devices	Survey scores and open comments	Opt-in text that people choose to share
Reads emotions in text	No	No	Looks for self-harm, suicide and violence risk	Groups comments into topics with a sentiment label	Seven emotional states
Explains each alert	Activity indicators only	Limited or not published	Shows the flagged activity	Topic titles and summaries	Words behind every result
Graded output	At risk or not	Risk scores	Severity levels	Engagement trends	Four risk levels and a 0 to 100 distress score
Privacy approach	Institutional data	Institutional data	Monitoring of school devices	Comments shown only for groups above a minimum size	Opt-in; managers see group trends only
Cost and access	Free, open source	Paid licence	Paid licence	Paid licence	Free plan, then low-cost Pro and Max plans

The comparison shows where the engine fits. Education analytics see activity rather than feelings, and school safety tools read text but work by monitoring devices. Survey tools read comments but summarise them by topic. The engine is built to give a graded and explained emotional state from text that people choose to share, and it sends people who need help towards counselling instead of replacing it.

8.2.3 SWOT analysis

The strengths, weaknesses, opportunities and threats of the engine are summarised in Table 8.3.

Table 8.3: SWOT analysis of the emotional burnout engine

Area	Points
Strengths	Explanation with every alert; seven graded states instead of a yes-or-no flag; base-size models that run on ordinary hardware; one engine for several markets; part of a multi-signal early-warning system; results tested statistically
Weaknesses	Trained on a public social-media corpus; English only; not yet validated with students, employees or counsellors; weaker on the rarest classes
Opportunities	Growing focus on student and employee well-being; burnout recognised as a workplace syndrome; data protection laws that favour transparent, consent-based tools; Sinhala and Tamil versions; Moodle and staff survey plugins
Threats	Mistrust of text analysis; misuse by employers; changes in regulation; large LMS and HR vendors adding their own AI features; harm from missed or false alerts

8.3 Business or sustainability model

8.3.1 Pricing plans: Free, Pro and Max

The engine is offered as an online service with three plans, in the same way as many software products. The Free plan lets a counsellor or a small team try the engine at no cost. The Pro plan is the main paid plan for one institution or company. The Max plan adds the full team platform, API access and local retraining for large clients. Prices are set per person covered per year, so the cost grows with the size of the client. Table 8.4 shows the three plans. All prices are indicative estimates for the Sri Lankan market and would be checked in pilots.

Table 8.4: Pricing plans of the emotional burnout engine (indicative prices)

Plan	Best for	What is included	Price per year (LKR)
Free	One counsellor, a small class or a small team trying the engine	Quick Analysis of typed text (up to 500 texts a month); seven emotional states, risk level, distress score and word heatmap; weekly check-ins for up to 100 people; email support	Free
Pro	A faculty, a private institute or a company	Everything in Free with no monthly limit; weekly progression for each person; class or team trends; critical-alert escalation; CSV and JSON export; Moodle or staff survey connector; onboarding workshop	Education: 100 per student. Workplaces and hospitals: 1,200 per person
Max	Universities, large companies, hospitals and app partners	Everything in Pro; fusion with the team's academic and behavioural signals on the R26-IT-059 platform; API access; yearly retraining on the client's own consented data; Sinhala and Tamil versions (planned); on-premise option; service agreement and priority support	Education: 250 per student (full team package). Workplaces and hospitals: 2,000 per person. API: 500 per 1,000 texts after the free monthly allowance

The Free plan is important for growth. It costs very little to run, because the engine works on ordinary processors, and it lets counsellors see the value before their institution pays. The Max price for education matches the team's full R26-IT-059 package, so a university that buys the complete early-warning system gets the emotional signal as part of it.

8.3.2 Business model canvas

Table 8.5 summarises the business model on the nine blocks of a business model canvas.

Table 8.5: Business model canvas of the emotional burnout engine

Block	Content
Customer segments	Universities and private institutes (first market); companies; hospitals; app developers through the API; online learning platforms
Value propositions	Early and explained emotional-risk signals from text that people choose to share; privacy-first design; low running cost
Channels	The Free plan; pilots at SLIIT; the R26-IT-059 platform; a Moodle plugin; staff survey connectors; API documentation
Customer relationships	Free plan first, then Pro or Max; onboarding workshops; a named support contact; a yearly model review with each client
Revenue streams	Pro and Max subscriptions per person per year; API usage fees; setup fees; training workshops; custom language versions
Key resources	Trained models and the risk mapping; the API and dashboard; the retraining pipeline; advice from qualified counsellors
Key activities	Retraining on consented local data; safety and fairness checks; product development; customer support
Key partners	SLIIT and other universities; counselling services and the national helpline; HR and LMS vendors; cloud providers
Cost structure	Cloud hosting; GPU time for retraining; developers; clinical safety review; sales, pilots and legal advice

8.3.3 Costs, break-even and revenue

Table 8.6 estimates the yearly cost of running the engine in its first year.

Table 8.6: Estimated yearly operating cost in the first year (indicative)

Cost item	Assumption	Yearly cost (LKR)
Cloud computing	A virtual machine that keeps the two transformer models in memory and scores text on a CPU	180,000
GPU time	Retraining and checking the models once a year on each client's consented data	60,000
Maintenance	Part-time developer for monitoring, updates and support	360,000
Clinical safety review	A qualified counsellor or psychologist checks labels, recommendations and escalation rules each year	150,000
Sales, pilots and compliance	Pilot workshops and legal advice on data protection	300,000
Total		1,050,000

On the Pro plan at LKR 100 per student, about 10,500 students cover this cost (1,050,000 ÷ 100). That is one large university or two to three private institutes. The cost stays low because the models are base-sized and run on ordinary processors; a graphics card is needed only for retraining.

Table 8.7 shows a cautious three-year scenario. It assumes an average of 5,000 students per institution, 500 employees per company and 1,000 staff per hospital, at the prices in Table 8.4. These are planning assumptions, not forecasts.

Table 8.7: Indicative three-year revenue scenario (LKR)

Item	Year 1	Year 2	Year 3
Education on Pro (2, 4 and 6 institutions)	1,000,000	2,000,000	3,000,000
Companies on Pro (0, 3 and 6 companies)	0	1,800,000	3,600,000
Hospitals on Max (0, 0 and 1 hospital)	0	0	2,000,000
API usage (0, 0.6 and 1.2 million paid texts)	0	300,000	600,000
Total revenue	1,000,000	4,100,000	9,200,000
Operating cost (grows with support needs)	1,050,000	1,600,000	2,500,000
Result	-50,000	2,500,000	6,700,000

In this scenario the engine almost covers its cost in the first year with education alone and makes a profit from the second year, when the first companies join. By the third

year, workplaces and hospitals bring in more than half of the revenue, which shows why the plan does not depend on one market.

8.3.4 Go-to-market roadmap

- **Phase 1 (months 0 to 6).** A Free plan pilot at SLIIT with one counselling unit, after ethics approval, using opt-in weekly reflections. The aim is to find out whether counsellors find the alerts and explanations useful.
- **Phase 2 (months 6 to 12).** Retrain and validate the models on the consented pilot data and publish the results. Launch the Pro plan for private institutes and state universities through the R26-IT-059 platform, with a Moodle plugin.
- **Phase 3 (year 2).** A workplace pilot with one IT or BPO company and its counselling partner, with team-level reports only. Add a staff survey connector and open the API to the first app partners on the Max plan.
- **Phase 4 (year 3).** A hospital pilot on the Max plan, Sinhala and Tamil models, and plugins for online learning platforms. Partnerships with counselling services make sure that every critical alert has a clear escalation route.

Sustainability. Economically, serving more than one market means that the product does not depend on one type of customer, and the subscriptions cover hosting, retraining and maintenance after the research project ends. Technically, the models are small, the code is documented and under version control, and a new developer can retrain the pipeline. Socially, the design keeps a person in charge of every decision and asks for consent, which protects trust over time.

8.4 Scalability intellectual property and compliance

Scalability. The prediction service keeps no state between requests, so more copies can be started behind a load balancer as demand grows, and weekly scoring can run as a batch job at night. Each client's data and retrained models are kept separate, so one engine can serve many institutions and companies. During evaluation, the ensemble classified a single passage in about 50 milliseconds on the project GPU.

Intellectual property. The code, the risk mapping and the dashboard are the author's own work within the team project. The pretrained models used, bert-base-uncased and roberta-base, are published under open licences that allow commercial

use. The research corpus is compiled from several public datasets with their own terms [49], so a commercial version should be retrained on each client's own consented data instead of shipping weights trained on the research corpus.

Compliance in Sri Lanka. The Personal Data Protection Act No. 9 of 2022 [72] is coming into operation in stages. Its main parts were first due to start on 18 March 2025, but that date was withdrawn [73]. The Act was then amended in 2025, and a later order set 1 January 2027 as the start date for the parts on processing personal data and on the duties of controllers and processors [74]. A product launched after that date must follow the Act from the first day. Data concerning health is a special category of personal data under the Act [75], and text about a person's emotional state should be treated in the same way. A deployment therefore needs informed consent that can be withdrawn, clear information about what is analysed and who sees the results, data minimisation, encryption with role-based access and audit logs, and respect for the rights of data subjects. In this design no decision is fully automated, because a person reviews every alert.

Clients in the European Union. The General Data Protection Regulation would apply, and health data is a special category under Article 9 [76]. The EU Artificial Intelligence Act bans AI systems that infer the emotions of people in workplaces and educational institutions, except for medical or safety reasons [77]. The Commission's guidelines explain that this ban covers systems based on biometric data, such as faces or voices, and that inferring emotions from written text is outside it [78]. However, AI systems used to monitor and evaluate the behaviour of workers are classed as high-risk [77], so a legal review would be needed before entering that market.

Rules for workplaces. Because an employer holds power over its staff, workplace clients follow stricter rules:

- employees opt in, and they can stop at any time without any effect on their job;
- only text written for well-being purposes is analysed, never work email or chat;
- managers see only team-level trends, and only for groups of at least ten people;
- an individual alert goes only to a confidential counsellor, and only with the employee's consent;

- the contract bans any use for hiring, firing, promotion, pay or performance reviews.

The main risks of a deployment and how they are reduced are listed in Table 8.8. The product must never be marketed as a diagnostic tool; it supports trained staff and does not replace professional judgement.

Table 8.8: Main deployment risks and how they are reduced

Risk	Possible effect	How it is reduced
A person at risk is missed	Harm and false reassurance	The engine is never the only safeguard; normal counselling channels stay open; recall is monitored
False alarm	Worry and stigma for the person	Human review before any contact; supportive wording
Privacy breach	Harm to people and penalties under data protection law	Encryption, role-based access, scores instead of raw text, limited retention
Misuse for grading or staff evaluation	Loss of trust and legal risk	Written usage policy, contract terms, team-level reports and audit logs
Lower accuracy on local writing	Unfair or wrong alerts	Retraining on consented local data each year; fairness checks; local-language models

8.5 Contribution to the Sustainable Development Goals

The component contributes mainly to SDG 3, Good Health and Well-being, and in particular to target 3.4, which includes promoting mental health and well-being [79], by helping staff notice emotional distress earlier. It also supports SDG 4, Quality Education, through target 4.3 on equal access to quality tertiary education [80], because students who receive support in time are more likely to stay in their studies. Through its use in workplaces and hospitals, it can also support SDG 8, Decent Work and Economic Growth, and its target 8.8 on safe and secure working environments for all workers [81].

These are intended contributions, and measuring the impact will need a real deployment and an evaluation. Possible negative effects include stigma for flagged people, a feeling of being watched, and unequal accuracy for different groups. To protect the benefits, use must be opt-in, alerts must stay confidential, every alert must be reviewed by a person, and fairness must be checked regularly.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This component reads student text, places it in one of seven emotional states, turns the result into a risk level and a distress score, and shows the words behind each prediction. A soft-voting ensemble of fine-tuned BERT and RoBERTa reached 82.32% accuracy and a macro F1 of 0.789, with the most balanced result across the rare classes among the ten models compared. The component runs as a REST API with an advisor dashboard and passes its score to the team's fusion layer.

9.2 Achievement of objectives

Table 9.1 summarises how each objective was achieved and where the evidence is reported.

Table 9.1: Achievement of objectives

Objective	Status	Evidence
O1 Clean seven-class corpus and fixed split	Achieved	Sections 3.2 and 3.3
O2 Ten models compared under one protocol	Achieved	Table 6.1
O3 BERT and RoBERTa soft-voting ensemble	Achieved	Table 6.2
O4 Class imbalance handled; weighted loss tested	Achieved	Table 6.5
O5 Word-level explanations checked with LIME and SHAP	Achieved	Figure 6.4
O6 Risk level, distress score and stress score	Achieved	Table 3.4
O7 Statistical significance of model differences	Achieved	Table 6.3; Table 6.4
O8 API, dashboard and team integration	Achieved	Sections 5.2 and 5.3

9.3 Individual research and product contributions

The research of this component was prepared for publication in two forms:

- An individual research paper, “Burnout Detection from Student Generated Text using Transformer Based NLP with Phrase Level Explainability”, was written and revised after review by the supervisors.
- A joint paper combining this component with the behavioural anomaly detection component (IT22215710) and the fusion layer was submitted to the International Conference on Advancements in Computing (ICAC 2026).

The product contributions are the Flask REST API, the Emotional Analysis dashboard and the score export used by the Meta-Integration layer.

9.4 Future directions

- Retrain and test the models on consented student reflections, with ethics approval.
- Follow the same consenting students across a semester to test the lead-time idea.
- Improve the rare classes with more examples and data augmentation.
- Compare domain-adapted and larger models, such as MentalRoBERTa [50], under the same protocol.
- Add Sinhala and Tamil support for Sri Lankan students.
- Run a study with counsellors to measure whether the explanations help their decisions.
- Pilot the engine in one company on the Pro plan, with team-level reports only (Section 8.3.4).

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GLOSSARY

Ablation study: An experiment that changes or removes one part of a method to measure its effect.

Attention (self-attention): The part of a transformer that lets each word weigh every other word in the text; its weights can be read as a sign of which words the model focused on.

Bootstrap confidence interval: A range for a result, found by re-sampling the test set many times with replacement and recomputing the result each time.

Distress score: A number from 0 to 100 that combines the seven class probabilities, weighted by how severe each class is.

Ensemble (soft voting): Combining several models by averaging their predicted probabilities.

Fine-tuning: Further training of a pretrained model on a specific task with labelled examples.

Lead time: The time between when a warning could be raised and the point at which it would be too late to help.

Macro F1: The F1 score averaged over the classes with equal weight for each class.

McNemar's test: A test that compares two models on the same test items by counting the items that only one of them got right.

Proxy dataset: A dataset used in place of the target data when that data is not available.

Stratified split: A split of the data in which each class keeps the same share in every part.

Weighted F1: The F1 score averaged over the classes, with each class weighted by its number of test items.

Zero-shot classification: Classifying text into labels the model was never trained on, here by asking whether a label statement follows from the text.

APPENDICES

Appendix A Extended results and sample outputs

Table A.1 gives the full per-class results of the class-weighted loss ablation, extending Table 6.5.

Table A.1: Per-class precision, recall and F1 of the plain and class-weighted RoBERTa models

Class	Plain P	Plain R	Plain F1	Weighted P	Weighted R	Weighted F1
Normal	0.956	0.950	0.953	0.960	0.921	0.940
Anxiety	0.845	0.893	0.868	0.856	0.864	0.860
Bipolar	0.837	0.819	0.828	0.795	0.829	0.812
Depression	0.787	0.752	0.769	0.818	0.666	0.735
Suicidal	0.708	0.748	0.727	0.660	0.805	0.725
Stress	0.684	0.750	0.716	0.588	0.826	0.687
Personality Disorder	0.698	0.552	0.617	0.590	0.612	0.601

Listing A.1 shows a shortened response of the analysis endpoint for the exam-stress text used in Chapter 5. The attention list is cut to the first six words.

Listing A.1: Shortened response of POST /api/analyze for the exam-stress text

```
{
  "prediction": "Stress",
  "confidence": 0.5939,
  "risk_level": "Medium",
  "stress_score": 40.6,
  "probabilities": {
    "Stress": 0.5939, "Normal": 0.267, "Depression": 0.0549,
    "Anxiety": 0.0414, "Suicidal": 0.0351,
    "Personality disorder": 0.0047, "Bipolar": 0.0029
  },
  "models": {
    "agree": true,
    "bert": { "prediction": "Stress", "confidence": 0.4339 },
    "roberta": { "prediction": "Stress", "confidence": 0.7538 }
  },
  "attention": [
    { "token": "i", "weight": 0.072671 },
    { "token": "am", "weight": 0.059895 },
    { "token": "so", "weight": 0.054626 },
    { "token": "stressed", "weight": 0.094164 },
    { "token": "about", "weight": 0.032696 },
    { "token": "my", "weight": 0.031466 }
  ],
  "model": "BERT + RoBERTa ensemble (soft voting)"
}
```

Appendix B Academic integrity and AI use disclosure

Generative AI was used during this project in accordance with the module policy, as follows.

- **Tool.** Claude (Anthropic), used through the Claude app and through Claude Code in the development environment.
- **Thesis preparation.** Claude helped to draft the structure and text of this thesis from the author's own research paper, notebooks, saved results files and dashboard screenshots, and to check grammar and IEEE reference formatting.
- **Checking.** Claude was used to check every reported number against the saved results files and notebook outputs, and to check each cited source against the original publication. This led to corrections of some values and model names used in the earlier paper, for example the weighted F1 of BERT and RoBERTa and the name of the emotion-pretrained model. It also led to the removal of sources that could not be confirmed.
- **Figures.** Claude drew Figure 2.1, Figure 4.2 and Figure 5.4 from the author's design, and redrew Figure 6.1 from the saved results.
- **Software.** Claude Code was used for debugging, for parts of the dashboard and API code, and to prepare the final version of the prediction service and the dashboard screenshots.

All models were trained and evaluated by the author with the author's own code and data. All AI-assisted text was reviewed, edited and verified by the author, who takes full responsibility for the content of this thesis.